CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

M & A DAIRY

14651 S CARLUCCI RD

DOS PALOS CA 93620

NOTES:

PREPARED BY:

CARDOSO AG SERVICES
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: M and A Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
14651 S Carlucci RD	Dos Palos	Merced	93620

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1930

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X085-X032-X007-XXXX X085-X032-X013-XXXX

B. OPERATORS

Goncalves, Manuel or Anthony

Operator name: Goncalves, Manuel or Anthony

Telephone no.: (209) 387-4313 Landline (209) 769-4336 Cellular

Mailing Address Number and Street	City	State	Zip Code
14651 Carlucci RD	Dos Palos	CA	93620

This operator is responsible for paying permit fees.

C. OWNERS

Goncalves, Manuel or Anthony

Legal owner name: Goncalves, Manuel or Anthony

Telephone no.: (209) 387-4313 Landline (209) 769-4336 Cellular

Mailing Address Number and Street	City	State	Zip Code
14651 Carlucci RD	Dos Palos	CA	93620

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	453	72	144	286	0	0
Number under roof	0	0	0	0	0	0
Maximum number	453	72	144	286	0	0
Average number	453	72	144	286	0	0
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 71 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 17,070.97 *tons per reporting period*

Total nitrogen from manure: 206,569.30 *lbs per reporting period*

After ammonia losses (30% loss applied): 144,598.51 *lbs per reporting period*

Total phosphorus from manure: 34,488.45 *lbs per reporting period*

Total potassium from manure: 87,051.30 *lbs per reporting period*

Total salt from manure: 229,851.45 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 16,852,800 *gallons*

Total nitrogen generated: 54,672.90 *lbs*

Total phosphorus generated: 13,677.31 *lbs*

Total potassium generated: 63,897.57 *lbs*

Total salt generated: 433,724.77 *lbs*

	16,852,800	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	16,852,800	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	35	35	2	process wastewater	X085-X032-X013-XXXX
10	14	14	2	manure	X085-X030-X003-XXXX
12	49	49	2	none	X085-X030-X003-XXXX
13	49	49	2	manure	X085-X030-X003-XXXX
16	33	33	2	none	X085-X029-X036-XXXX
17	34	34	2	none	X085-X029-X036-XXXX
18	24	24	2	manure	X085-X029-X036-XXXX
19	30	30	2	manure	X085-X029-X036-XXXX
2	47	47	2	none	X085-X032-X003-XXXX
20	26	26	2	none	X085-X029-X012-XXXX X085-X030-X004-XXXX
21	33	33	2	manure	X085-X029-X012-XXXX X085-X030-X004-XXXX
22	28	28	2	manure	X085-X030-X009-XXXX
23	50	50	2	manure	X085-X030-X009-XXXX X085-X032-X013-XXXX
24	53	53	2	manure	X085-X032-X007-XXXX
25	47	47	2	process wastewater	X085-X032-X007-XXXX
5	23	23	2	manure	X085-X030-X003-XXXX
Totals for areas that were used for application	386	386	22		
Totals for areas that were not used for application	189	189	10		
Land application area totals	575	575	32		

B. CROPS AND HARVESTS

1

Field name: 1

1

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	489.00 *ton*	Dry-weight		40.6	12,000.00	1,900.00	10,700.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.97	199.18	31.54	177.60	1,643.21

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	992.00 *ton*	Dry-weight		60.9	13,000.00	3,000.00	11,700.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.34	288.13	66.49	259.32	1,440.67

10

Field name: 10

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	196.00 *ton*	Dry-weight		65.2	13,000.00	2,800.00	16,200.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	126.67	27.28	157.85	1,023.12

10

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	420.00 *ton*	Dry-weight		76.1	9,300.00	2,600.00	27,200.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	133.36	37.28	390.05	1,878.54

12

Field name: 12

10/22/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 49 Plant date: 10/22/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	346.00 *ton*	Dry-weight		34.9	28,800.00	3,300.00	30,300.00		9.30
08/02/2023	346.00 *ton*	Dry-weight		34.9	28,800.00	3,300.00	30,300.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	14.12	529.56	60.68	557.14	1,710.03

13

Field name: 13

13

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 49 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	720.00 *ton*	Dry-weight		65.2	13,000.00	2,800.00	16,200.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.69	132.95	28.64	165.68	1,073.83

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 49 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,344.00 *ton*	Dry-weight		76.1	9,300.00	2,600.00	27,200.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.43	121.93	34.09	356.62	1,717.52

16

Field name: 16

10/05/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 33 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	231.00 *ton*	Dry-weight		41.6	34,900.00	3,200.00	28,500.00		10.40
08/02/2023	154.00 *ton*	Dry-weight		2.4	34,100.00	3,100.00	27,600.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	11.67	595.97	54.40	484.43	1,642.82

17

Field name: 17

10/05/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 34 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	231.00 *ton*	Dry-weight		5.6	31,300.00	3,100.00	32,300.00		11.10
08/02/2023	154.00 *ton*	Dry-weight		3.0	26,900.00	2,900.00	35,300.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	11.32	637.87	65.25	724.50	2,390.41

18

Field name: 18

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	448.00 *ton*	Dry-weight		41.2	9,000.00	2,900.00	14,100.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	24.00	132.80	0.00
Total actual harvest content	18.67	197.57	63.66	309.52	1,865.92

18

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/17/2023	896.00 *ton*	Dry-weight		70.0	14,300.00	3,000.00	9,200.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	37.33	320.32	67.20	206.08	1,590.40

19

Field name: 19

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	480.00 *ton*	Dry-weight		41.2	9,000.00	2,900.00	14,100.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	169.34	54.57	265.31	1,599.36

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/17/2023	1,024.00 *ton*	Dry-weight		70.0	14,300.00	3,000.00	9,200.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.13	292.86	61.44	188.42	1,454.08

2

Field name: 2

10/30/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 47 Plant date: 10/30/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	352.00 *ton*	Dry-weight		34.9	28,800.00	3,300.00	30,300.00		9.30
08/02/2023	352.00 *ton*	Dry-weight		34.9	28,800.00	3,300.00	30,300.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	14.98	561.67	64.36	590.92	1,813.71

20

Field name: 20

11/03/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 26 Plant date: 11/03/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	157.00 *ton*	Dry-weight		31.7	32,300.00	3,400.00	17,300.00		8.82
08/02/2023	157.00 *ton*	Dry-weight		31.7	32,300.00	3,400.00	17,300.00		8.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	12.08	532.86	56.09	285.40	1,455.04

21

Field name: 21

21

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	448.00 *ton*	Dry-weight		41.2	9,000.00	2,900.00	14,100.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.58	143.69	46.30	225.11	1,357.03

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/17/2023	960.00 *ton*	Dry-weight		70.0	14,300.00	3,000.00	9,200.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.09	249.60	52.36	160.58	1,239.27

22

Field name: 22

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	392.00 *ton*	Dry-weight		59.3	15,200.00	2,800.00	17,600.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	173.22	31.91	200.57	1,162.39

22

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	840.00 *ton*	Dry-weight		75.3	10,900.00	2,500.00	24,900.00		14.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	161.54	37.05	369.02	2,208.18

23

Field name: 23

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	988.00 *ton*	Dry-weight		42.8	11,200.00	3,100.00	14,500.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.76	253.18	70.08	327.78	1,944.07

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,716.00 *ton*	Dry-weight		60.0	10,700.00	2,800.00	12,700.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.32	293.78	76.88	348.69	1,702.27

24

Field name: 24

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 53 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	918.00 *ton*	Dry-weight		59.3	15,200.00	2,800.00	17,600.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.32	214.31	39.48	248.14	1,438.11

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 53 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,632.00 *ton*	Dry-weight		75.2	10,900.00	2,500.00	24,900.00		14.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.79	166.48	38.18	380.30	2,275.69

25

Field name: 25

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 47 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	768.00 *ton*	Dry-weight		42.8	11,200.00	3,100.00	14,500.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.34	209.37	57.95	271.05	1,607.64

25

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 47 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,536.00 *ton*	Dry-weight		60.0	10,700.00	2,800.00	12,700.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.68	279.75	73.21	332.04	1,620.97

5

Field name: 5

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 23 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	336.00 *ton*	Dry-weight		40.6	12,000.00	1,900.00	1,070.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.61	208.26	32.97	18.57	1,718.16

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	720.00 *ton*	Dry-weight		60.9	13,000.00	3,000.00	11,700.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.30	318.24	73.44	286.42	1,591.20

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/05/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.41	14.49	110.58	590.61	878,400.00 *gal*
Application event totals		98.41	14.49	110.58	590.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.95	18.18	91.14	755.34	1,056,000.00 *gal*
Canal	Surface water	0.17	0.00	0.00	0.00	7,038,951.00 *gal*
Application event totals		85.12	18.18	91.14	755.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	88.88	23.77	108.01	604.27	1,056,000.00 *gal*
Canal	Surface water	0.08	0.00	0.00	0.00	3,512,306.00 *gal*
Application event totals		88.96	23.77	108.01	604.27	

1 - 05/30/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.55	25.82	117.34	656.46	1,147,200.00 *gal*
Canal	Surface water	0.11	0.00	0.00	0.00	4,593,738.00 *gal*
Application event totals		96.66	25.82	117.34	656.46	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.55	25.82	117.34	656.46	1,147,200.00 *gal*
Canal	Surface water	0.28	0.00	0.00	0.00	11,594,682.00 *gal*
Application event totals		96.83	25.82	117.34	656.46	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.67	29.76	119.02	1,013.99	960,000.00 *gal*
Canal	Surface water	0.14	0.00	0.00	0.00	5,944,521.00 *gal*
Application event totals		101.82	29.76	119.02	1,013.99	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.22	0.00	0.00	0.00	9,256,353.00 *gal*
Application event totals		0.22	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.11	0.00	0.00	0.00	4,818,260.00 *gal*
Application event totals		0.11	0.00	0.00	0.00	

1 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.67	29.76	119.02	1,013.99	960,000.00 *gal*
Canal	Surface water	0.12	0.00	0.00	0.00	4,927,538.00 *gal*
Application event totals		101.79	29.76	119.02	1,013.99	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	2,852,785.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

10 - 11/22/2022: Oats, silage-soft dough

Field name: 10

Crop: Oats, silage-soft dough Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	169.77	52.97	266.78	0.00	153.00 *ton*
Application event totals		169.77	52.97	266.78	0.00	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	2,986,222.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	1,490,069.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

10 - 05/30/2023: Corn, silage

Field name: 10

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	178.72	56.87	251.83	0.00	255.00 *ton*
Application event totals		178.72	56.87	251.83	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	1,948,859.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	4,918,956.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	2,521,918.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	3,926,938.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

10 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	2,044,110.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	2,090,471.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	1,210,273.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

12 - 10/22/2021: Alfalfa, hay

Field name: 12

Crop: Alfalfa, hay

Plant date: 10/22/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	5,215,242.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	6,821,005.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

12 - 10/22/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	17,216,347.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	8,826,713.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	13,744,282.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	7,154,385.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	7,316,647.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	4,235,954.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

13 - 10/05/2022: Oats, silage-soft dough

13 - 10/05/2022: Oats, silage-soft dough

Field name: 13

Crop: Oats, silage-soft dough

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	175.32	54.70	275.50	0.00	553.00 *ton*
Application event totals		175.32	54.70	275.50	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	10,451,776.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	5,215,242.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

13 - 05/30/2023: Corn, silage

Field name: 13

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	162.60	51.74	229.12	0.00	812.00 *ton*
Application event totals		162.60	51.74	229.12	0.00	

13 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	6,821,005.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	17,216,347.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	8,826,713.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	13,744,282.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	7,154,385.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	7,316,647.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

13 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	4,235,954.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

16 - 10/05/2022: Alfalfa, hay

Field name: 16

Crop: Alfalfa, hay

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	7,038,951.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	3,512,306.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	4,593,738.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	5,797,341.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

16 - 10/05/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	5,797,341.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	5,944,521.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	9,256,353.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

17 - 10/05/2022: Alfalfa, hay

Field name: 17

Crop: Alfalfa, hay

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	7,252,253.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	3,618,739.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

17 - 10/05/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	4,732,943.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	5,973,018.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	5,973,018.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	6,124,658.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	9,536,848.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

18 - 10/05/2022: Oats, silage-soft dough

Field name: 18

Crop: Oats, silage-soft dough

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

18 - 10/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	261.50	81.59	410.92	0.00	404.00 *ton*
Application event totals		261.50	81.59	410.92	0.00	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	5,119,237.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	2,554,404.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

18 - 05/30/2023: Corn, silage

Field name: 18

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	252.66	80.39	356.02	0.00	618.00 *ton*
Application event totals		252.66	80.39	356.02	0.00	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	3,340,901.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

18 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	8,432,496.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	4,323,288.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	6,731,893.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	3,504,189.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	3,583,664.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Subsurface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	2,074,753.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

19 - 10/05/2022: Oats, silage-soft dough

19 - 10/05/2022: Oats, silage-soft dough

Field name: 19

Crop: Oats, silage-soft dough

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	229.91	71.74	361.29	0.00	444.00 *ton*
Application event totals		229.91	71.74	361.29	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	6,399,047.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	3,193,005.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

19 - 05/30/2023: Corn, silage

Field name: 19

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	252.82	80.44	356.25	0.00	773.00 *ton*
Application event totals		252.82	80.44	356.25	0.00	

19 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	4,176,126.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	10,540,620.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	5,404,110.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	8,414,866.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	4,380,236.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	4,479,580.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

19 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	2,593,441.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

2 - 10/30/2019: Alfalfa, hay

Field name: 2

Crop: Alfalfa, hay

Plant date: 10/30/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	5,002,375.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	6,542,597.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	16,513,639.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	8,466,439.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

2 - 10/30/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	13,183,290.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	6,862,370.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	7,018,009.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	4,063,058.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

20 - 11/03/2018: Alfalfa, hay

Field name: 20

Crop: Alfalfa, hay

Plant date: 11/03/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	2,767,271.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

20 - 11/03/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	3,619,309.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	9,135,204.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	4,683,562.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	7,292,884.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	3,796,204.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	3,882,303.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

20 - 11/03/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	2,247,649.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

21 - 11/22/2022: Oats, silage-soft dough

Field name: 21

Crop: Oats, silage-soft dough Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	193.94	60.52	304.77	0.00	412.00 *ton*
Application event totals		193.94	60.52	304.77	0.00	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	3,512,306.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

21 - 05/30/2023: Corn, silage

Field name: 21

Crop: Corn, silage Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

21 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	303.28	96.50	427.35	0.00	1,020.00 *ton*
Application event totals		303.28	96.50	427.35	0.00	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	4,593,738.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	11,594,682.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	5,944,521.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	9,256,353.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	4,818,260.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

21 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	4,927,538.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	2,852,785.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

22 - 11/22/2022: Oats, silage-soft dough

Field name: 22

Crop: Oats, silage-soft dough Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	234.13	73.05	367.91	0.00	422.00 *ton*
Application event totals		234.13	73.05	367.91	0.00	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	2,980,138.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

22 - 05/30/2023: Corn, silage

Field name: 22

Crop: Corn, silage Plant date: 05/30/2023

22 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	217.62	69.24	306.64	0.00	621.00 *ton*
Application event totals		217.62	69.24	306.64	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	3,897,717.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	9,837,912.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	5,043,836.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	7,853,875.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	4,088,220.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

22 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	4,180,941.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	2,420,545.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

23 - 10/05/2022: Oats, silage-soft dough

Field name: 23

Crop: Oats, silage-soft dough

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	346.73	108.19	544.86	0.00	1,116.00 *ton*
Application event totals		346.73	108.19	544.86	0.00	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	10,665,078.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	5,321,675.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

23 - 05/30/2023: Corn, silage

Field name: 23

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	303.39	96.53	427.50	0.00	1,546.00 *ton*
Application event totals		303.39	96.53	427.50	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	6,960,210.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	17,567,701.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	9,006,850.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	14,024,777.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	7,300,393.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

23 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	7,465,967.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	4,322,402.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

24 - 10/05/2022: Oats, silage-soft dough

Field name: 24

Crop: Oats, silage-soft dough

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	292.52	91.27	459.67	0.00	998.00 *ton*
Application event totals		292.52	91.27	459.67	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	11,304,982.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	5,640,976.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

24 - 05/30/2023: Corn, silage

Field name: 24

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	224.75	71.51	316.69	0.00	1,214.00 *ton*
Application event totals		224.75	71.51	316.69	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	7,377,822.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	18,621,763.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	9,547,261.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	14,866,264.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	7,738,417.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

24 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	7,913,925.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	4,581,746.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

25 - 10/05/2022: Oats, silage-soft dough

Field name: 25

Crop: Oats, silage-soft dough Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.11	14.15	108.00	576.81	1,152,000.00 *gal*
Application event totals		96.11	14.15	108.00	576.81	

03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.77	20.92	104.90	869.30	1,632,000.00 *gal*
Canal	Surface water	0.18	0.00	0.00	0.00	10,025,173.00 *gal*
Application event totals		97.95	20.92	104.90	869.30	

25 - 10/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.25	24.14	109.69	613.62	1,440,000.00 *gal*
Canal	Surface water	0.09	0.00	0.00	0.00	5,002,375.00 *gal*
Application event totals		90.34	24.14	109.69	613.62	

25 - 05/30/2023: Corn, silage

Field name: 25

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.25	24.14	109.69	613.62	1,440,000.00 *gal*
Canal	Surface water	0.12	0.00	0.00	0.00	6,542,597.00 *gal*
Application event totals		90.37	24.14	109.69	613.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.25	24.14	109.69	613.62	1,440,000.00 *gal*
Canal	Surface water	0.29	0.00	0.00	0.00	16,513,639.00 *gal*
Application event totals		90.55	24.14	109.69	613.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.32	29.36	117.44	1,000.51	1,272,000.00 *gal*
Canal	Surface water	0.15	0.00	0.00	0.00	8,466,439.00 *gal*
Application event totals		100.47	29.36	117.44	1,000.51	

25 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	13,183,290.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	6,862,370.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.32	29.36	117.44	1,000.51	1,272,000.00 *gal*
Canal	Surface water	0.12	0.00	0.00	0.00	7,018,009.00 *gal*
Application event totals		100.45	29.36	117.44	1,000.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	4,063,058.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

5 - 10/05/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	282.32	88.09	443.65	0.00	418.00 *ton*
Application event totals		282.32	88.09	443.65	0.00	

5 - 10/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	2,447,971.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

5 - 05/30/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	252.98	80.49	356.47	0.00	593.00 *ton*
Application event totals		252.98	80.49	356.47	0.00	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	3,201,696.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	8,081,142.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	4,143,151.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

5 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	6,451,397.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	3,358,181.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	3,434,345.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

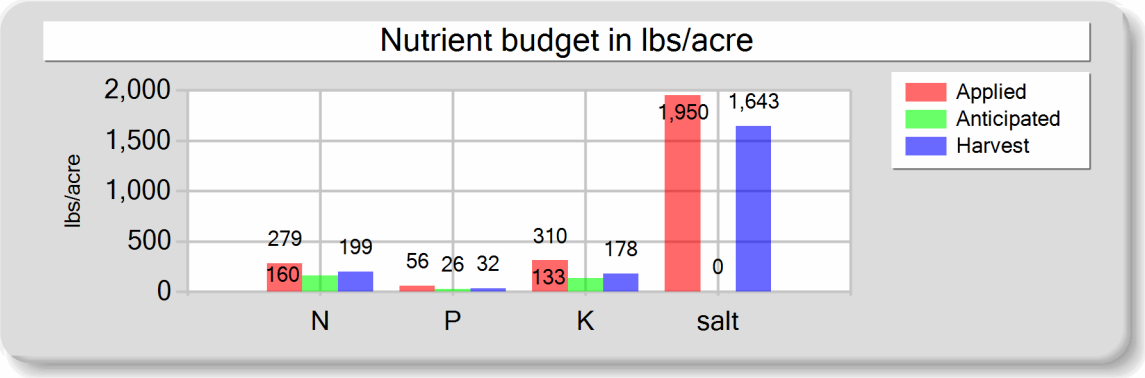
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	1,988,305.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 10/05/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	272.24	56.44	309.74	1,950.22
Fresh water	0.25	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.49	56.44	309.74	1,950.22
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	199.18	31.54	177.60	1,643.21
Nutrient balance	80.32	24.90	132.14	307.02
Applied to removed ratio	1.40	1.79	1.74	1.19

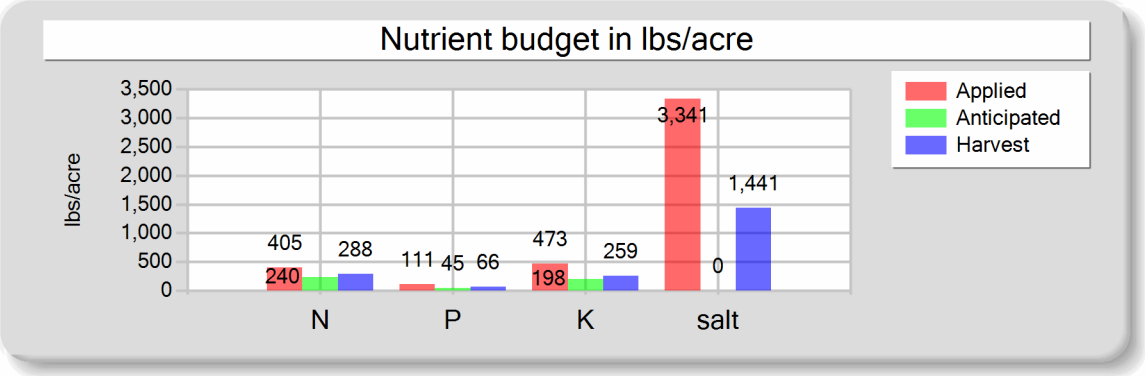
Fresh water applied	
10,551,257.00	*gallons*
388.57	*acre-inches*
11.10	*inches/acre*

Process wastewater applied	
2,990,400.00	*gallons*
110.13	*acre-inches*
3.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 05/30/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	396.46	111.15	472.73	3,340.90
Fresh water	1.05	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	404.50	111.15	472.73	3,340.90
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	288.13	66.49	259.32	1,440.67
Nutrient balance	116.37	44.66	213.41	1,900.23
Applied to removed ratio	1.40	1.67	1.82	2.32

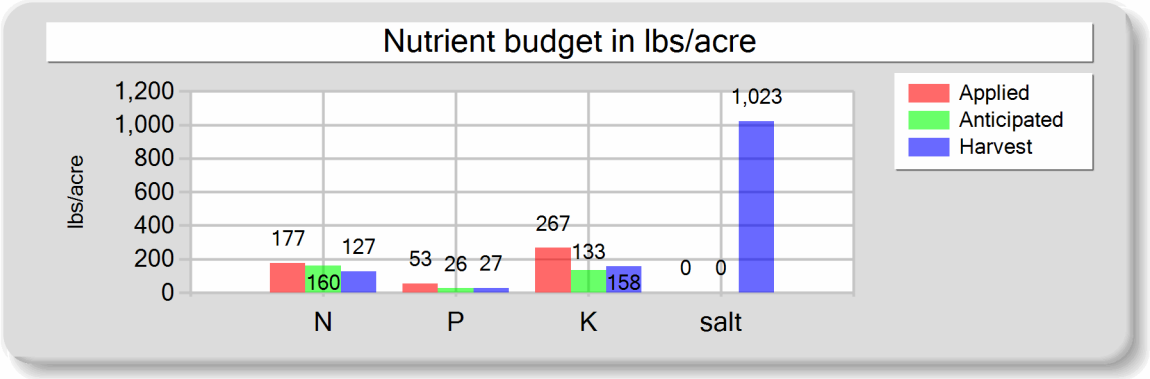
Fresh water applied	
43,987,877.00	*gallons*
1,619.92	*acre-inches*
46.28	*inches/acre*

Process wastewater applied	
4,214,400.00	*gallons*
155.20	*acre-inches*
4.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 11/22/2022: Oats, silage-soft dough

Field name: 10 Crop: Oats, silage-soft dough Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	169.77	52.97	266.78	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	177.04	52.97	266.78	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	126.67	27.28	157.85	1,023.12
Nutrient balance	50.36	25.69	108.93	-1,023.12
Applied to removed ratio	1.40	1.94	1.69	0.00

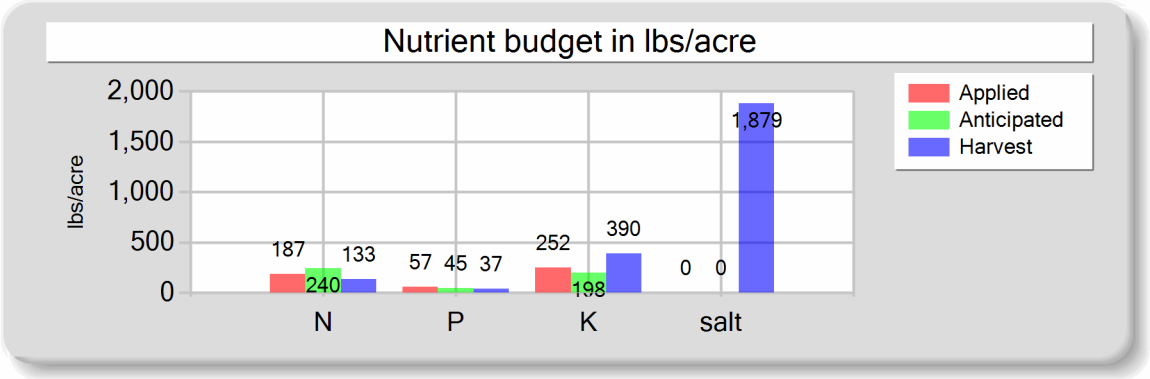
Fresh water applied	
4,476,291.00	*gallons*
164.85	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 05/30/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.72	56.87	251.83	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	186.83	56.87	251.83	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	133.36	37.28	390.05	1,878.54
Nutrient balance	53.47	19.58	-138.22	-1,878.54
Applied to removed ratio	1.40	1.53	0.65	0.00

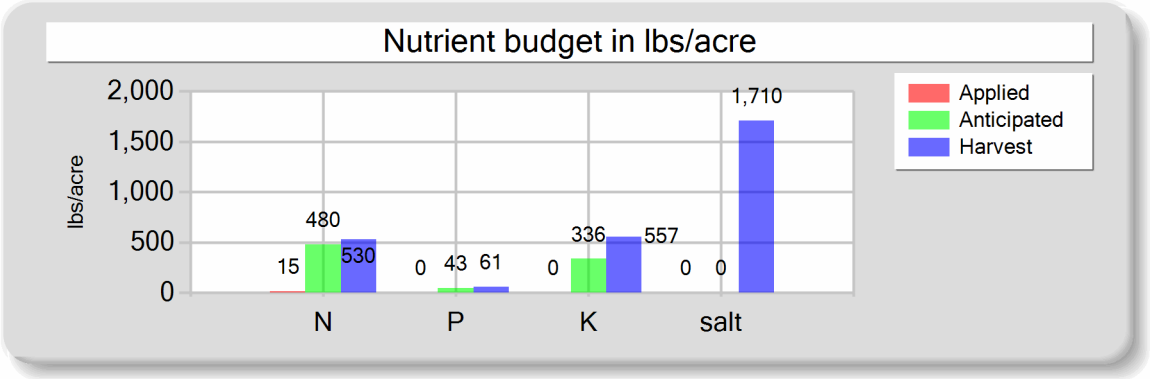
Fresh water applied	
18,661,525.00	*gallons*
687.24	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12 - 10/22/2021: Alfalfa, hay

Field name: 12 Crop: Alfalfa, hay Plant date: 10/22/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.20	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	15.20	0.00	0.00	0.00
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	529.56	60.68	557.14	1,710.03
Nutrient balance	-514.36	-60.68	-557.14	-1,710.03
Applied to removed ratio	0.03	0.00	0.00	0.00

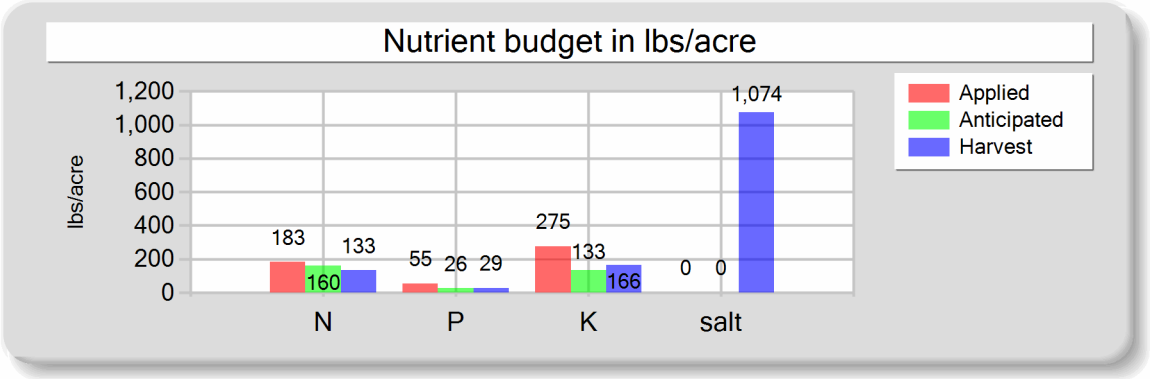
Fresh water applied	
70,530,575.00	*gallons*
2,597.40	*acre-inches*
53.01	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
2	*harvests*

13 - 10/05/2022: Oats, silage-soft dough

Field name: 13 Crop: Oats, silage-soft dough Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	175.32	54.70	275.50	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	182.58	54.70	275.50	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	132.95	28.64	165.68	1,073.83
Nutrient balance	49.63	26.07	109.82	-1,073.83
Applied to removed ratio	1.37	1.91	1.66	0.00

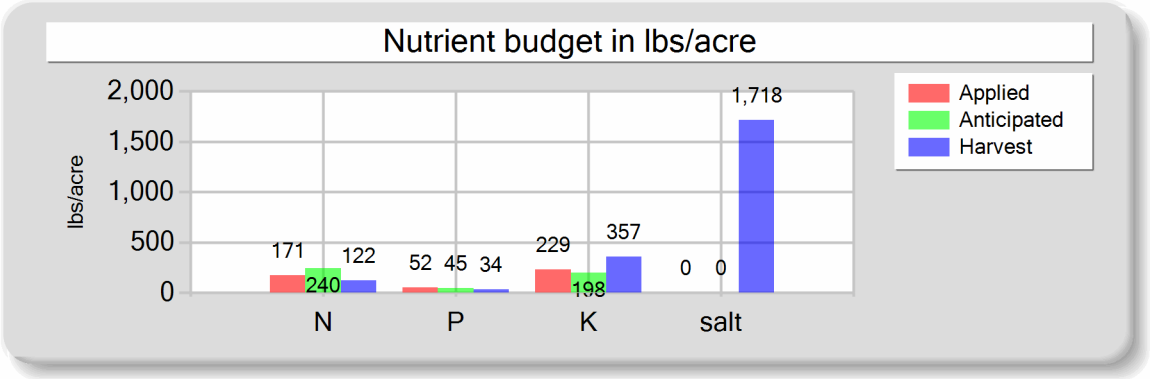
Fresh water applied	
15,667,018.00	*gallons*
576.96	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

13 - 05/30/2023: Corn, silage

Field name: 13 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	162.60	51.74	229.12	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.71	51.74	229.12	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	121.93	34.09	356.62	1,717.52
Nutrient balance	48.78	17.65	-127.50	-1,717.52
Applied to removed ratio	1.40	1.52	0.64	0.00

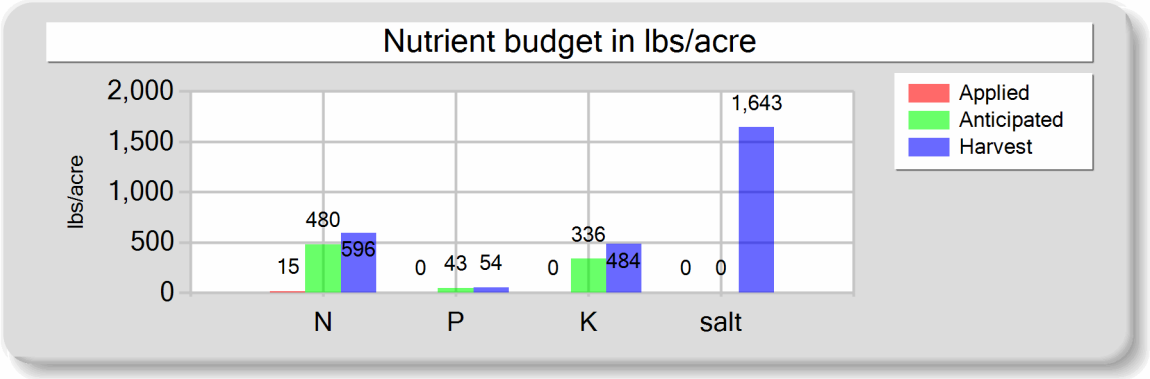
Fresh water applied	
65,315,333.00	*gallons*
2,405.34	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

16 - 10/05/2022: Alfalfa, hay

Field name: 16 Crop: Alfalfa, hay Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.06	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	15.06	0.00	0.00	0.00
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	595.97	54.40	484.43	1,642.82
Nutrient balance	-580.91	-54.40	-484.43	-1,642.82
Applied to removed ratio	0.03	0.00	0.00	0.00

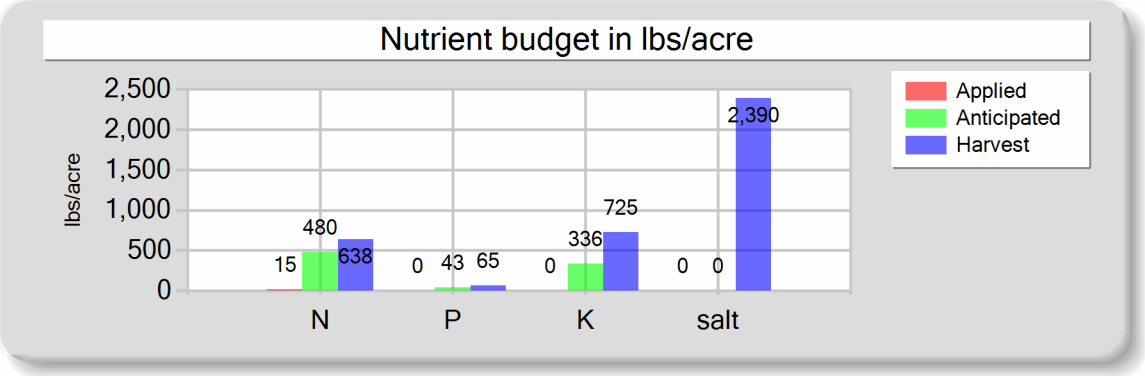
Fresh water applied	
41,940,551.00	*gallons*
1,544.53	*acre-inches*
46.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
2	*harvests*

17 - 10/05/2022: Alfalfa, hay

Field name: 17 Crop: Alfalfa, hay Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.06	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	15.06	0.00	0.00	0.00
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	637.87	65.25	724.50	2,390.41
Nutrient balance	-622.81	-65.25	-724.50	-2,390.41
Applied to removed ratio	0.02	0.00	0.00	0.00

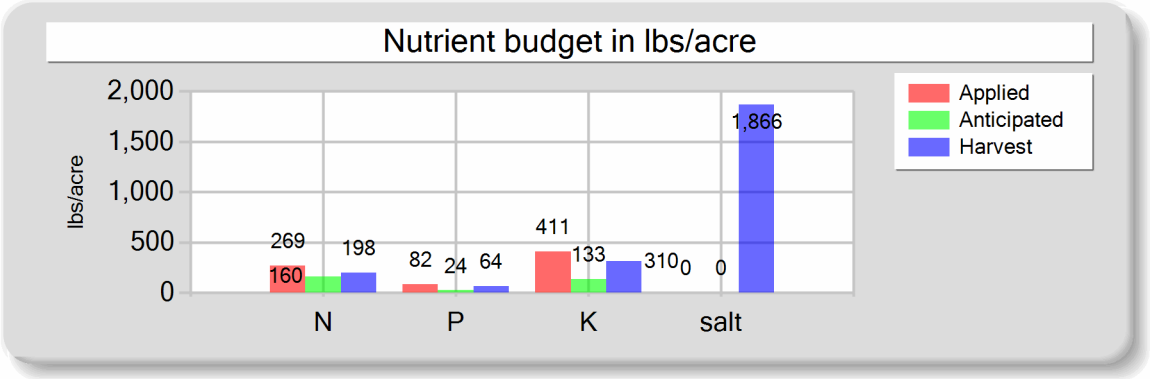
Fresh water applied	
43,211,477.00	*gallons*
1,591.33	*acre-inches*
46.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
2	*harvests*

18 - 10/05/2022: Oats, silage-soft dough

Field name: 18 Crop: Oats, silage-soft dough Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	261.50	81.59	410.92	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.76	81.59	410.92	0.00
Anticipated crop nutrient removal	160.00	24.00	132.80	0.00
Actual crop nutrient removal	197.57	63.66	309.52	1,865.92
Nutrient balance	71.19	17.93	101.40	-1,865.92
Applied to removed ratio	1.36	1.28	1.33	0.00

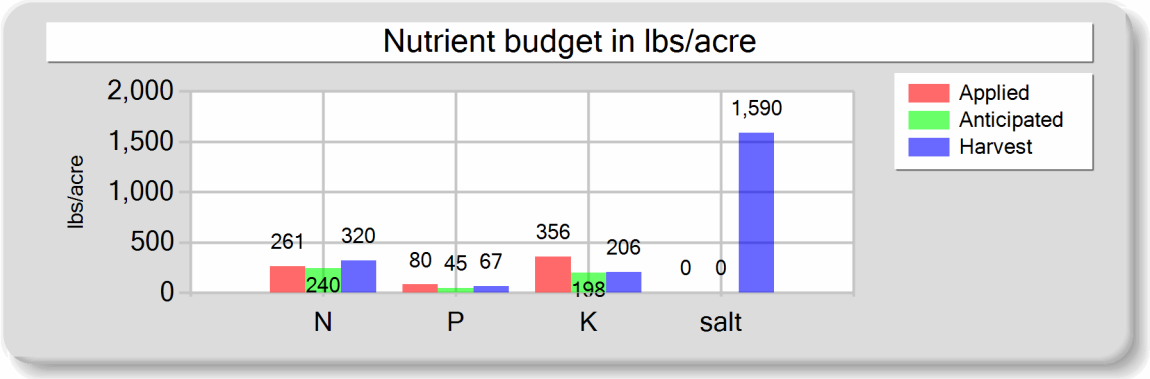
Fresh water applied	
7,673,641.00	*gallons*
282.59	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

18 - 05/30/2023: Corn, silage

Field name: 18 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	252.66	80.39	356.02	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	260.77	80.39	356.02	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	320.32	67.20	206.08	1,590.40
Nutrient balance	-59.55	13.19	149.94	-1,590.40
Applied to removed ratio	0.81	1.20	1.73	0.00

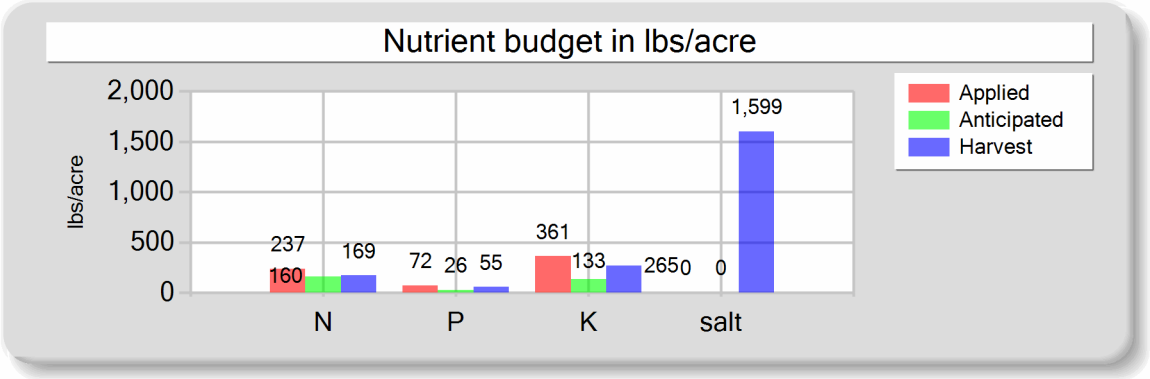
Fresh water applied	
31,991,184.00	*gallons*
1,178.13	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

19 - 10/05/2022: Oats, silage-soft dough

Field name: 19 Crop: Oats, silage-soft dough Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	229.91	71.74	361.29	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	237.18	71.74	361.29	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	169.34	54.57	265.31	1,599.36
Nutrient balance	67.83	17.17	95.98	-1,599.36
Applied to removed ratio	1.40	1.31	1.36	0.00

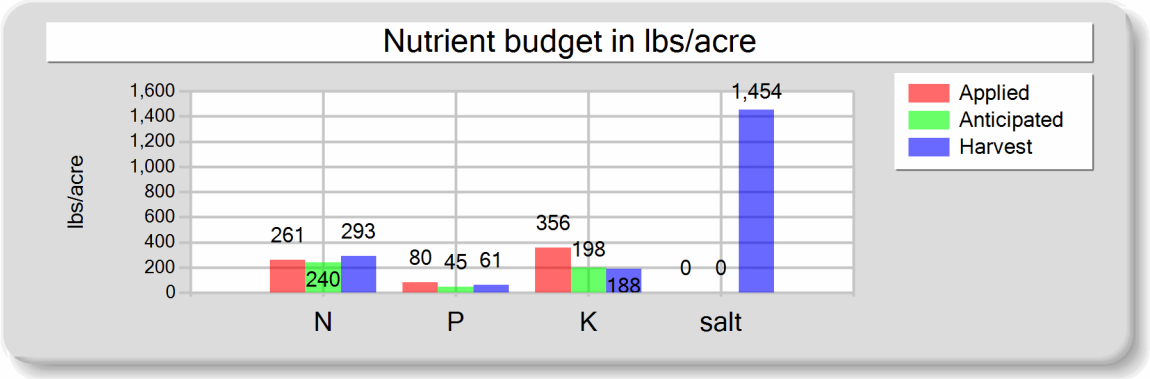
Fresh water applied	
9,592,052.00	*gallons*
353.24	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

19 - 05/30/2023: Corn, silage

Field name: 19 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	252.82	80.44	356.25	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	260.93	80.44	356.25	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	292.86	61.44	188.42	1,454.08
Nutrient balance	-31.93	19.00	167.83	-1,454.08
Applied to removed ratio	0.89	1.31	1.89	0.00

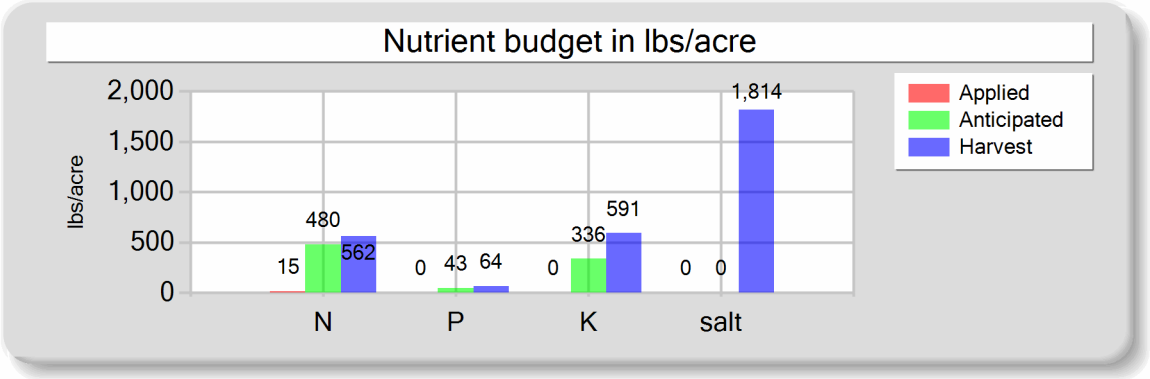
Fresh water applied	
39,988,979.00	*gallons*
1,472.66	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 10/30/2019: Alfalfa, hay

Field name: 2 Crop: Alfalfa, hay Plant date: 10/30/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.20	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	15.20	0.00	0.00	0.00
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	561.67	64.36	590.92	1,813.71
Nutrient balance	-546.46	-64.36	-590.92	-1,813.71
Applied to removed ratio	0.03	0.00	0.00	0.00

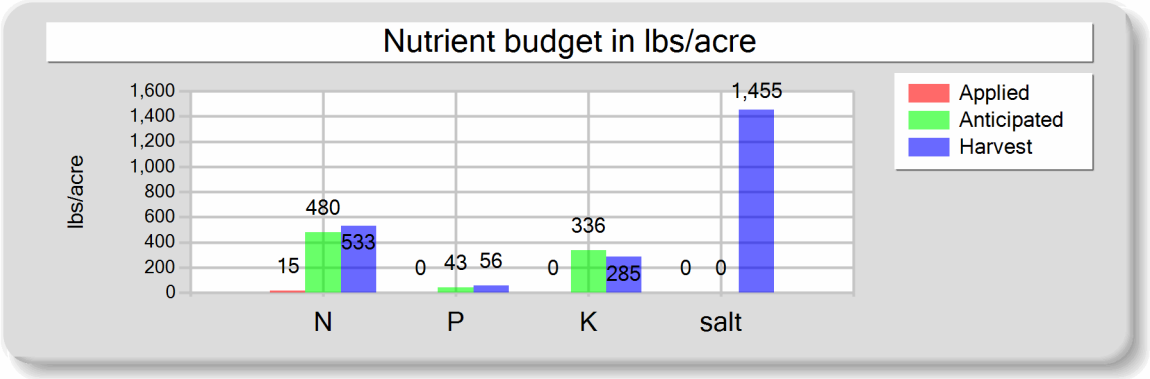
Fresh water applied	
67,651,777.00	*gallons*
2,491.38	*acre-inches*
53.01	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
2	*harvests*

20 - 11/03/2018: Alfalfa, hay

Field name: 20 Crop: Alfalfa, hay Plant date: 11/03/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.20	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	15.20	0.00	0.00	0.00
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	532.86	56.09	285.40	1,455.04
Nutrient balance	-517.65	-56.09	-285.40	-1,455.04
Applied to removed ratio	0.03	0.00	0.00	0.00

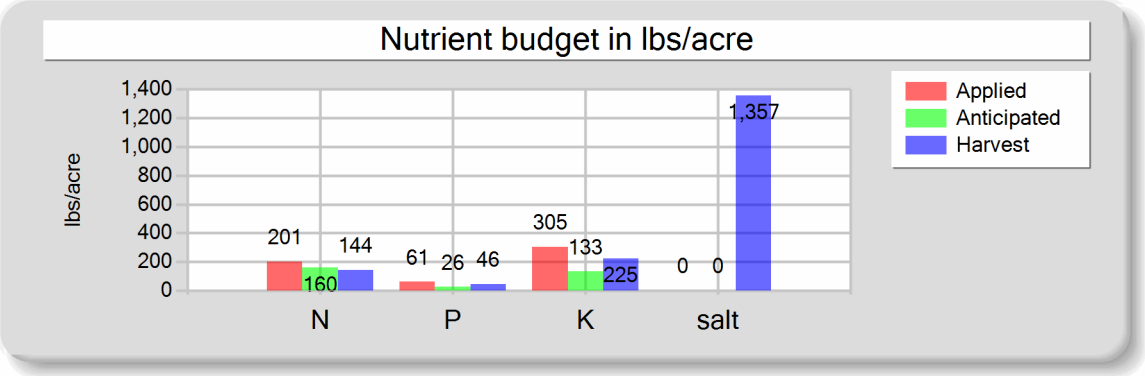
Fresh water applied	
37,424,386.00	*gallons*
1,378.21	*acre-inches*
53.01	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
2	*harvests*

21 - 11/22/2022: Oats, silage-soft dough

Field name: 21 Crop: Oats, silage-soft dough Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	193.94	60.52	304.77	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.09	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.03	60.52	304.77	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	143.69	46.30	225.11	1,357.03
Nutrient balance	57.35	14.22	79.66	-1,357.03
Applied to removed ratio	1.40	1.31	1.35	0.00

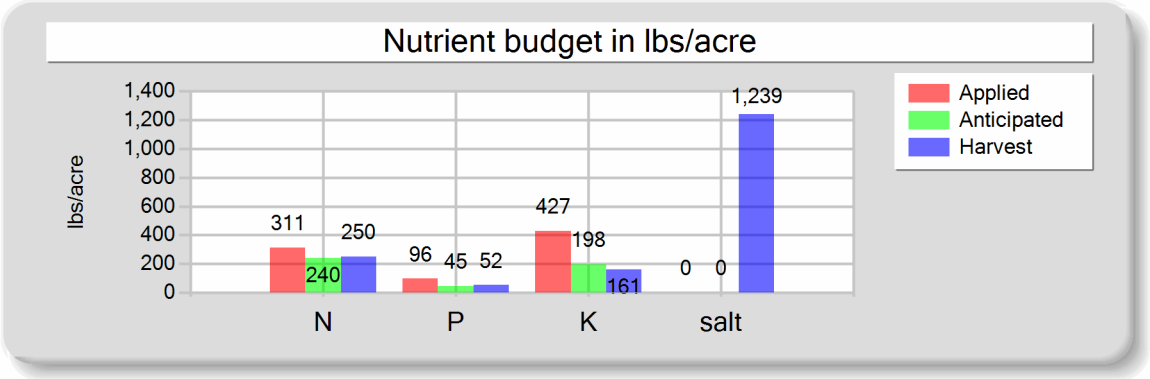
Fresh water applied
3,512,306.00 *gallons*
129.35 *acre-inches*
3.92 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

21 - 05/30/2023: Corn, silage

Field name: 21 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	303.28	96.50	427.35	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	311.39	96.50	427.35	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	249.60	52.36	160.58	1,239.27
Nutrient balance	61.79	44.13	266.77	-1,239.27
Applied to removed ratio	1.25	1.84	2.66	0.00

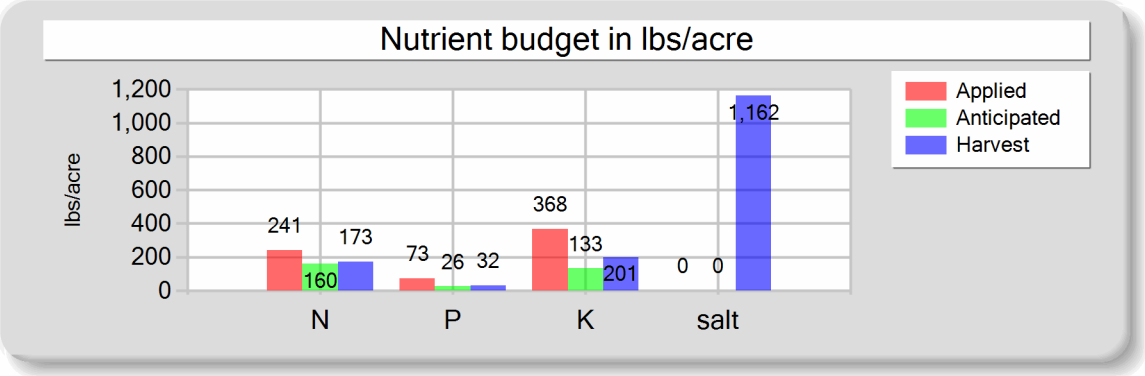
Fresh water applied	
43,987,877.00	*gallons*
1,619.92	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

22 - 11/22/2022: Oats, silage-soft dough

Field name: 22 Crop: Oats, silage-soft dough Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	234.13	73.05	367.91	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.09	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.21	73.05	367.91	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	173.22	31.91	200.57	1,162.39
Nutrient balance	68.00	41.15	167.34	-1,162.39
Applied to removed ratio	1.39	2.29	1.83	0.00

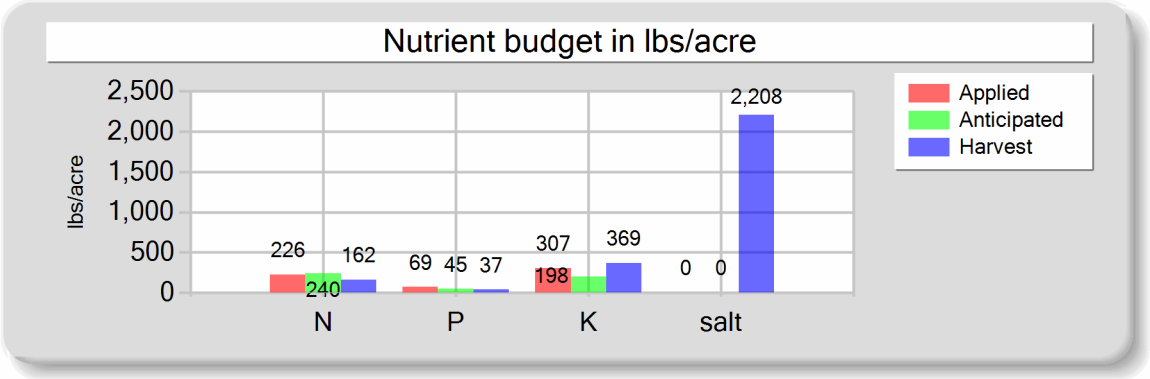
Fresh water applied	
2,980,138.00	*gallons*
109.75	*acre-inches*
3.92	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

22 - 05/30/2023: Corn, silage

Field name: 22 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	217.62	69.24	306.64	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.73	69.24	306.64	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	161.54	37.05	369.02	2,208.18
Nutrient balance	64.19	32.19	-62.38	-2,208.18
Applied to removed ratio	1.40	1.87	0.83	0.00

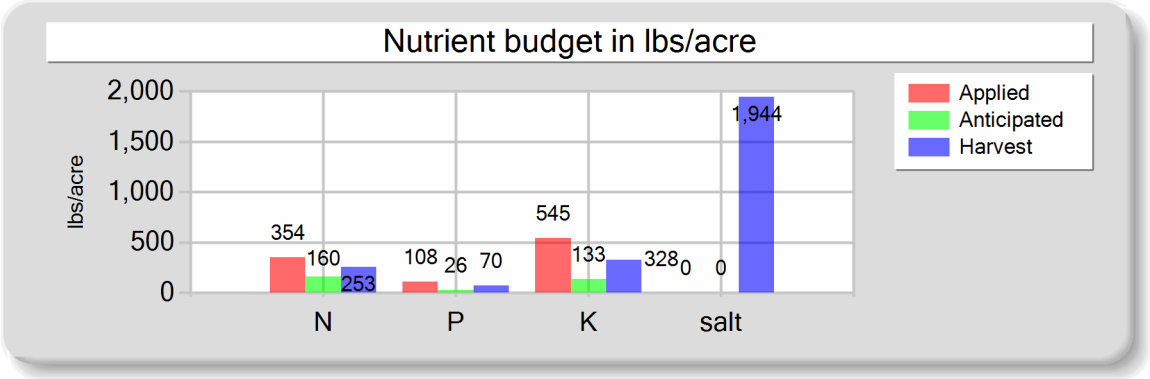
Fresh water applied	
37,323,046.00	*gallons*
1,374.48	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

23 - 10/05/2022: Oats, silage-soft dough

Field name: 23 Crop: Oats, silage-soft dough Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	346.73	108.19	544.86	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	353.99	108.19	544.86	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	253.18	70.08	327.78	1,944.07
Nutrient balance	100.81	38.11	217.08	-1,944.07
Applied to removed ratio	1.40	1.54	1.66	0.00

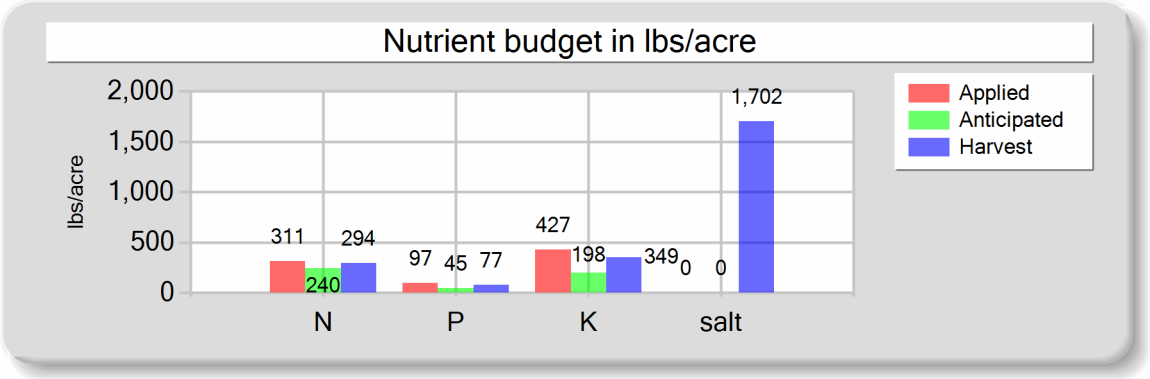
Fresh water applied	
15,986,753.00	*gallons*
588.74	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

23 - 05/30/2023: Corn, silage

Field name: 23 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	303.39	96.53	427.50	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	311.50	96.53	427.50	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	293.78	76.88	348.69	1,702.27
Nutrient balance	17.72	19.66	78.81	-1,702.27
Applied to removed ratio	1.06	1.26	1.23	0.00

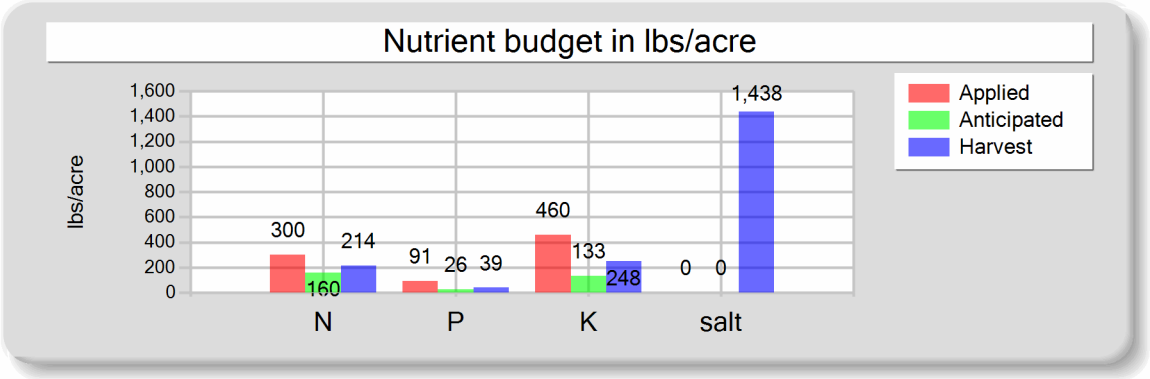
Fresh water applied	
66,648,300.00	*gallons*
2,454.43	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

24 - 10/05/2022: Oats, silage-soft dough

Field name: 24 Crop: Oats, silage-soft dough Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	292.52	91.27	459.67	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.78	91.27	459.67	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	214.31	39.48	248.14	1,438.11
Nutrient balance	85.48	51.80	211.52	-1,438.11
Applied to removed ratio	1.40	2.31	1.85	0.00

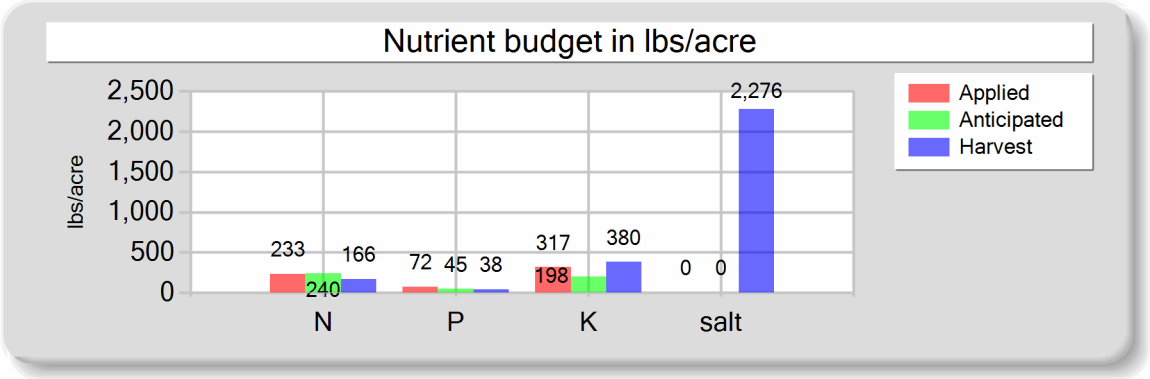
Fresh water applied	
16,945,958.00	*gallons*
624.06	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

24 - 05/30/2023: Corn, silage

Field name: 24 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	224.75	71.51	316.69	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.86	71.51	316.69	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	166.48	38.18	380.30	2,275.69
Nutrient balance	66.39	33.33	-63.61	-2,275.69
Applied to removed ratio	1.40	1.87	0.83	0.00

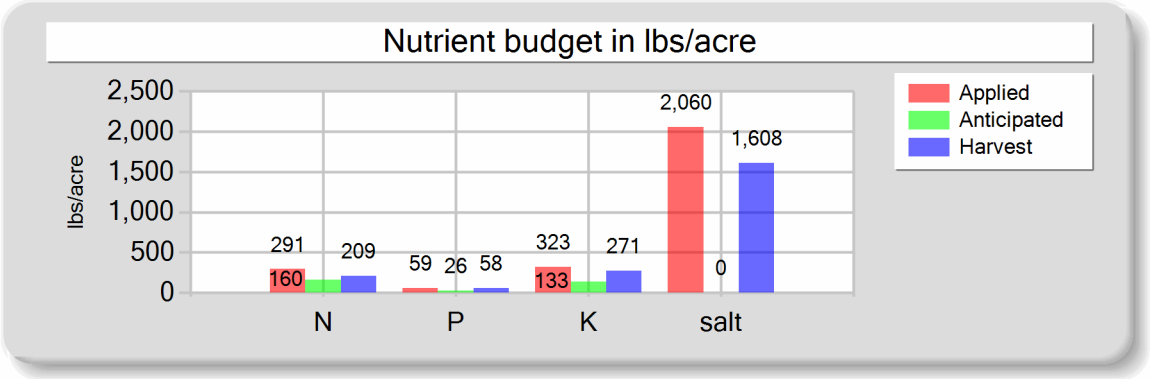
Fresh water applied	
70,647,198.00	*gallons*
2,601.70	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

25 - 10/05/2022: Oats, silage-soft dough

Field name: 25 Crop: Oats, silage-soft dough Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	284.14	59.21	322.58	2,059.73
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.40	59.21	322.58	2,059.73
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	209.37	57.95	271.05	1,607.64
Nutrient balance	82.04	1.26	51.52	452.09
Applied to removed ratio	1.39	1.02	1.19	1.28

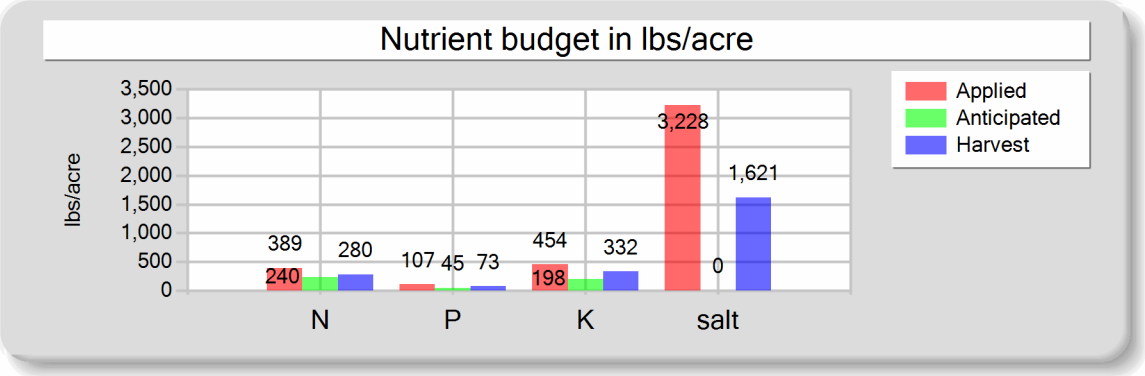
Fresh water applied	
15,027,548.00	*gallons*
553.41	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
4,224,000.00	*gallons*
155.56	*acre-inches*
3.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

25 - 05/30/2023: Corn, silage

Field name: 25 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	381.15	106.99	454.25	3,228.26
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	389.26	106.99	454.25	3,228.26
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	279.75	73.21	332.04	1,620.97
Nutrient balance	109.52	33.79	122.21	1,607.29
Applied to removed ratio	1.39	1.46	1.37	1.99

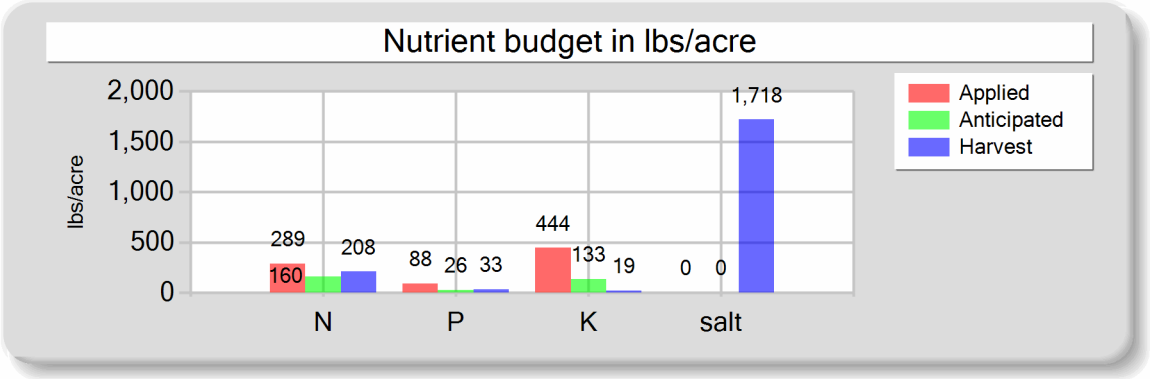
Fresh water applied	
62,649,402.00	*gallons*
2,307.16	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
5,424,000.00	*gallons*
199.75	*acre-inches*
4.25	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 10/05/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	282.32	88.09	443.65	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.09	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.41	88.09	443.65	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	208.26	32.97	18.57	1,718.16
Nutrient balance	81.15	55.12	425.08	-1,718.16
Applied to removed ratio	1.39	2.67	23.89	0.00

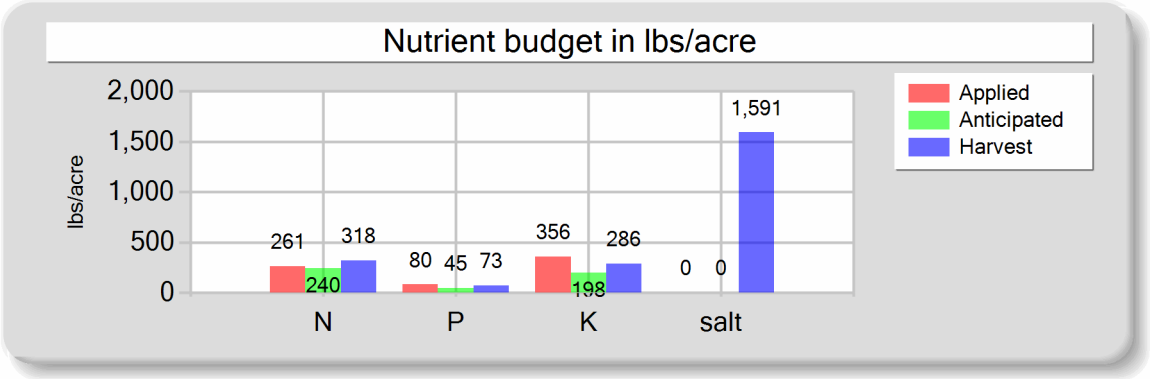
Fresh water applied
2,447,971.00 *gallons*
90.15 *acre-inches*
3.92 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

5 - 05/30/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	252.98	80.49	356.47	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.09	80.49	356.47	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	318.24	73.44	286.42	1,591.20
Nutrient balance	-57.15	7.05	70.05	-1,591.20
Applied to removed ratio	0.82	1.10	1.24	0.00

Fresh water applied	
30,658,217.00	*gallons*
1,129.04	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Corral Manure

Sample and source description: 2022 Corral Manure

Sample date: 10/10/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,600.00	8,300.00	41,800.00							0.00
DL	0.01	0.01	0.01							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 09/06/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,000.00	7,000.00	31,000.00							0.00
DL	0.01	0.01	0.01							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 12/13/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,900.00	4,700.00	18,700.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 2022

Sample and source description: Lagoon Q4 2022

Sample date: 11/30/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	469.00	332.00	0.00	0.90	69.20	528.00								5,970.00	2,820
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	337.00	90.00	0.00	0.40	72.20	362.00								3,220.00	3,000
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	352.00	227.00	0.00	1.00	94.40	429.00								5,250.00	2,400
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 08/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	443.00	351.00	0.00	1.20	130.00	520.00								6,330.00	4,430
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Slurry

Sample and source description: Slurry

Sample date: 09/18/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,710.00	254.00	0.00	0.50	392.00	583.00								6,830.00	3,150
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	608.00	400.00	0.00	0.70	119.00	451.00								6,420.00	3,000
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

San Luis Canal

Sample description: San Luis Canal

Sample date: 08/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.10	0.00	0.10								213.00	0
DL	0.01	0.01	0.01								10.00	1

Domestic Well

Domestic Well

DW MB

Sample description: DW MB

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.10	0.00	0.10	42.80	15.80	130.00	188.00	0.00	77.40	135.00	992.00	600
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	10.00	1

DW Shop

Sample description: DW Shop

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.20	0.00	0.20	45.10	16.60	136.00	186.00	0.00	82.70	150.00	977.00	530
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 10/05/2022: Oats, silage-soft dough

Field 1 Oats

Sample and source description: Field 1 Oats

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 40.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	1,900.00	10,700.00		9.90
DL	0.01	0.01	0.01		1.00

1 - 05/30/2023: Corn, silage

Field 1 Corn

Sample and source description: Field 1 Corn

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	3,000.00	11,700.00		6.50
DL	0.01	1.00	0.01		1.00

10 - 11/22/2022: Oats, silage-soft dough

Field 10 Oats

Sample and source description: Field 10 Oats

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,800.00	16,200.00		10.50
DL	0.01	0.01	0.01		1.00

10 - 05/30/2023: Corn, silage

Field 10 Corn Silage

Sample and source description: Field 10 Corn Silage

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,300.00	2,600.00	27,200.00		13.10
DL	0.01	0.01	0.01		1.00

12 - 10/22/2021: Alfalfa, hay

Field 12 Alfalfa

Sample and source description: Field 12 Alfalfa

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 34.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,800.00	3,300.00	30,300.00		9.30
DL	0.01	0.01	0.01		1.00

13 - 10/05/2022: Oats, silage-soft dough

Field 13 Oats

Sample and source description: Field 13 Oats

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,800.00	16,200.00		10.50
DL	0.01	0.01	0.01		1.00

13 - 05/30/2023: Corn, silage

Field 13 Corn Silage

Sample and source description: Field 13 Corn Silage

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,300.00	2,600.00	27,200.00		13.10
DL	0.01	0.01	0.01		1.00

16 - 10/05/2022: Alfalfa, hay

Field 16 Alfalfa

Sample and source description: Field 16 Alfalfa

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,900.00	3,200.00	28,500.00		10.40
DL	0.01	0.01	0.01		1.00

Field 16 Alfalfa

Sample and source description: Field 16 Alfalfa

Sample date: 08/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 2.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,100.00	3,100.00	27,600.00		8.70
DL	0.01	0.01	0.01		1.00

17 - 10/05/2022: Alfalfa, hay

Field 17 Alfalfa

Sample and source description: Field 17 Alfalfa

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 5.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,300.00	3,100.00	32,300.00		11.10
DL	0.01	0.01	0.01		1.00

17 - 10/05/2022: Alfalfa, hay

Field 17 Alfalfa

Sample and source description: Field 17 Alfalfa

Sample date: 08/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 3.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,900.00	2,900.00	35,300.00		11.00
DL	0.01	0.01	0.01		1.00

18 - 10/05/2022: Oats, silage-soft dough

Oats Field 18

Sample and source description: Oats Field 18

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,000.00	2,900.00	14,100.00		8.50
DL	0.01	0.01	0.01		1.00

18 - 05/30/2023: Corn, silage

Field 18 Corn

Sample and source description: Field 18 Corn

Sample date: 08/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	3,000.00	9,200.00		7.10
DL	0.01	0.01	0.01		1.00

19 - 10/05/2022: Oats, silage-soft dough

Oats Field 19

Sample and source description: Oats Field 19

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,000.00	2,900.00	14,100.00		8.50
DL	0.01	0.01	0.01		1.00

19 - 05/30/2023: Corn, silage

Field 19 Corn

Sample and source description: Field 19 Corn

Sample date: 08/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	3,000.00	9,200.00		7.10
DL	0.01	0.01	0.01		1.00

2 - 10/30/2019: Alfalfa, hay

Field 2 Alfalfa

Sample and source description: Field 2 Alfalfa

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 34.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,800.00	3,300.00	30,300.00		9.30
DL	0.01	0.01	0.01		1.00

20 - 11/03/2018: Alfalfa, hay

Field 20 Alfalfa

Sample and source description: Field 20 Alfalfa

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 31.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,300.00	3,400.00	17,300.00		8.82
DL	0.01	0.01	0.01		1.00

21 - 11/22/2022: Oats, silage-soft dough

Field 21 Oat Silage

Sample and source description: Field 21 Oat Silage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,000.00	2,900.00	14,100.00		8.50
DL	0.01	0.01	0.01		1.00

21 - 05/30/2023: Corn, silage

Field 21 Corn

Sample and source description: Field 21 Corn

Sample date: 08/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	3,000.00	9,200.00		7.10
DL	0.01	0.01	0.01		1.00

22 - 11/22/2022: Oats, silage-soft dough

Field 22 Oat Silage

Sample and source description: Field 22 Oat Silage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	2,800.00	17,600.00		10.20
DL	0.01	0.01	0.01		1.00

22 - 05/30/2023: Corn, silage

Field 22 Corn Silage

Sample and source description: Field 22 Corn Silage

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	2,500.00	24,900.00		14.90
DL	0.01	0.01	0.01		1.00

23 - 10/05/2022: Oats, silage-soft dough

Oats Field 23

Sample and source description: Oats Field 23

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	3,100.00	14,500.00		8.60
DL	0.01	0.01	0.01		1.00

23 - 05/30/2023: Corn, silage

Field 23 Corn Silage

Sample and source description: Field 23 Corn Silage

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	2,800.00	12,700.00		6.20
DL	0.01	0.01	0.01		1.00

24 - 10/05/2022: Oats, silage-soft dough

Field 24 Oat Silage

Sample and source description: Field 24 Oat Silage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	2,800.00	17,600.00		10.20
DL	0.01	0.01	0.01		1.00

24 - 05/30/2023: Corn, silage

Field 24 Corn Silage

Sample and source description: Field 24 Corn Silage

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	2,500.00	24,900.00		14.90
DL	0.01	0.01	0.01		1.00

25 - 10/05/2022: Oats, silage-soft dough

Oats Field 25

Sample and source description: Oats Field 25

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	3,100.00	14,500.00		8.60
DL	0.01	0.01	0.01		1.00

25 - 05/30/2023: Corn, silage

Field 25 Corn Silage

Sample and source description: Field 25 Corn Silage

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	2,800.00	12,700.00		6.20
DL	0.01	0.01	0.01		1.00

5 - 10/05/2022: Oats, silage-soft dough

Field 5 Oat Silage

Sample and source description: Field 5 Oat Silage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 40.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	1,900.00	1,070.00		9.90
DL	0.01	0.01	0.01		1.00

5 - 05/30/2023: Corn, silage

Field 5 Corn Silage

Sample and source description: Field 5 Corn Silage

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	3,000.00	11,700.00		6.50
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

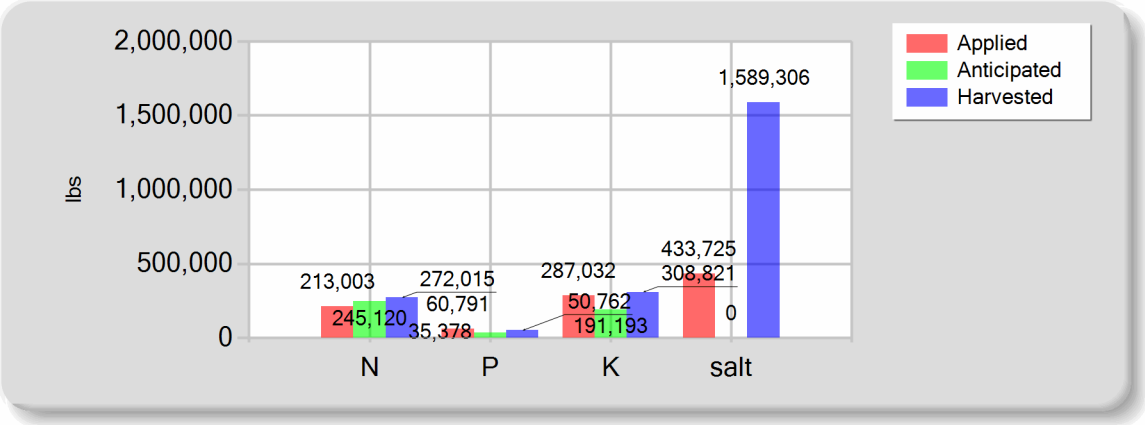
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

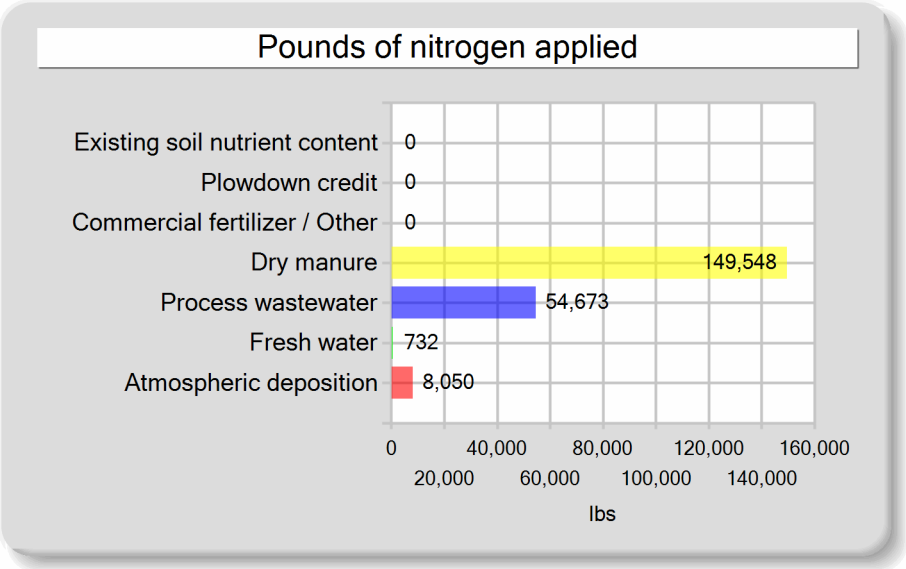
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

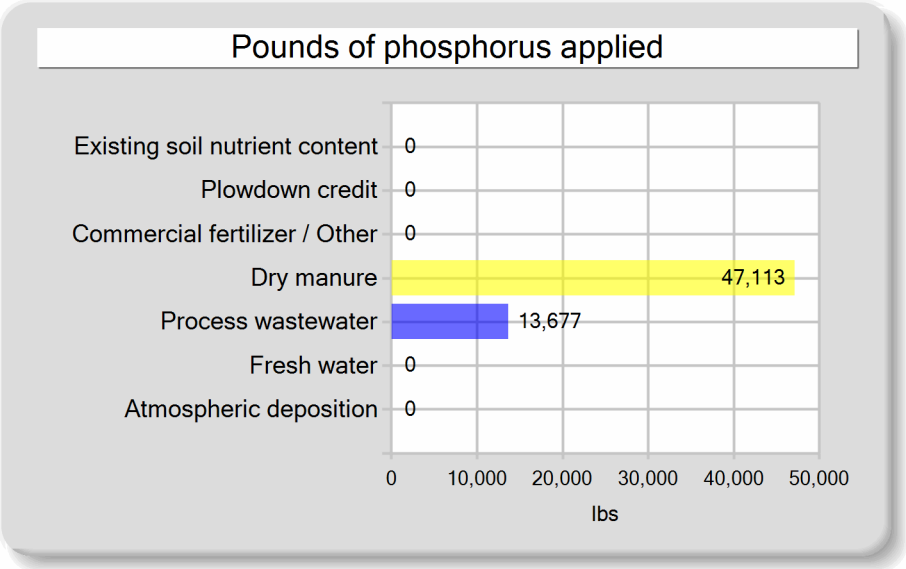
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	149,548.27	47,113.37	223,134.46	0.00
Process wastewater	54,672.90	13,677.31	63,897.57	433,724.77
Fresh water	732.26	0.00	0.00	0.00
Atmospheric deposition	8,050.00	0.00	0.00	0.00
Total nutrients applied	213,003.43	60,790.68	287,032.03	433,724.77
Anticipated crop nutrient removal	245,120.00	35,378.00	191,192.80	0.00
Actual crop nutrient removal	272,015.48	50,761.51	308,821.32	1,589,305.69
Nutrient balance	-59,012.05	10,029.17	-21,789.30	-1,155,580.93
Applied to removed ratio	0.78	1.20	0.93	0.27

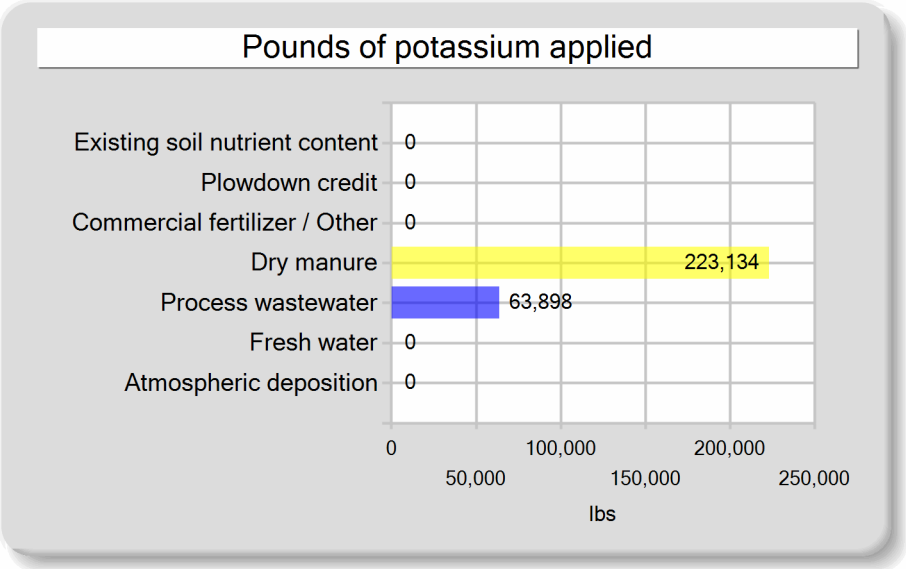
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

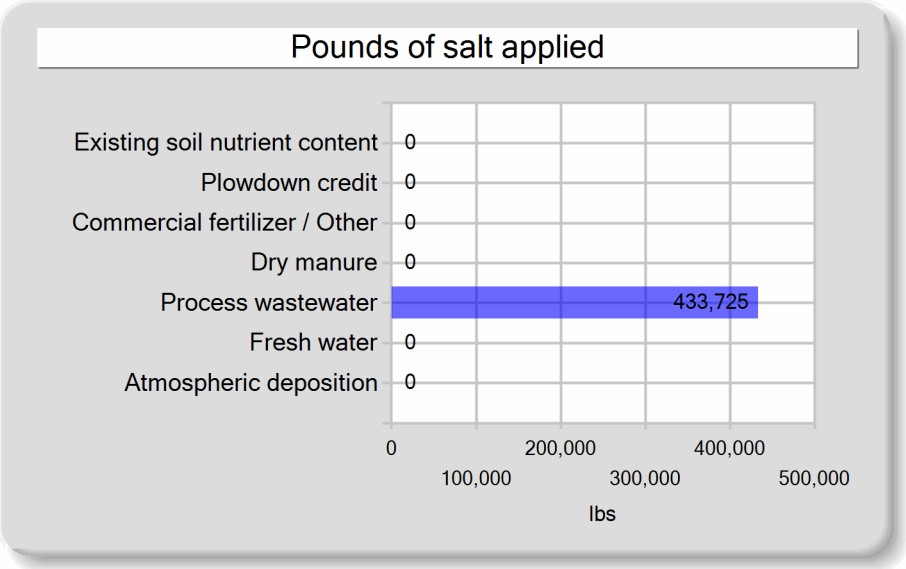


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Anthony Goncalves

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Manuel or Anthony Goncalves	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 16, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

M & A Dairy #1
14720 Carlucci Rd
Dos Palos CA 93620
00-0021357 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1014
Sampled By: M Pedroso
Requested By: Aurelia Goncalves
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	TDS mg/L	CO_3 as $CaCO_3$ mg/L	HCO_3 as $CaCO_3$ mg/L	Cl mg/L	SO_4 mg/L	Ca mg/L	Mg mg/L	Na mg/L	pH at 25°C unit
1	DW - Shop	12/13/23	14:30	977	0.98	0.2	ND		530	ND	186	150	82.7	45.1	16.6	136	7.9
2	DW - MB	12/13/23	14:35	992	0.99	0.1	ND		600	ND	188	135	77.4	42.8	15.8	130	7.8

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

M & A Dairy #1
14720 Carlucci Rd
Dos Palos CA 93620
00-0021357 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1014
Sampled By: M Pedroso
Requested By: Aurelia Goncalves
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - Shop	12/13/23	14:30	**977**	**0.98**	0.2	0.55	0.05
2	DW - MB	12/13/23	14:35	**992**	**0.99**	0.1	0.27	0.02

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L1014-01	DW - Shop	Ag Water	M Pedroso		12/13/2023 14:30
23L1014-02	DW - MB	Ag Water	M Pedroso		12/13/2023 14:35

Default Cooler Temperature on Receipt °C: 4.7
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Sample Results

Sample: DW - Shop
23L1014-01 (Water)

Sampled: 12/13/2023 14:30
Sampled By: M Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Alkalinity as CaCO3	**186**	mg/L	10.0	1		12/20/23 11:04	SM 2320 B		BEL0726
Calcium	**45.1**	mg/L	0.1	1		12/19/23 11:40	EPA 200.7		BEL0695
Chloride	**150**	mg/L	0.2	1	250	12/17/23 18:39	EPA 300.0		BEL0662
Carbonate as CaCO3	ND	mg/L	1	1		12/20/23 11:04	SM 2320 B		BEL0726
Electrical Conductivity	**0.98**	mmhos/cm	0.01	1		12/20/23 11:04	SM 2510 B		BEL0726
Electrical Conductivity umhos	**977**	umhos/cm	10.0	1		12/20/23 11:04	SM 2510 B		BEL0726
Bicarbonate as CaCO3	**186**	mg/L	5.00	1		12/20/23 11:04	SM 2320 B		BEL0726
Potassium	**2.14**	mg/L	0.500	1		12/19/23 11:40	EPA 200.7		BEL0695
Magnesium	**16.6**	mg/L	0.1	1		12/19/23 11:40	EPA 200.7		BEL0695
Sodium	**136**	mg/L	1	1		12/19/23 11:40	EPA 200.7		BEL0695
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 14:30	Field		BEL0654
Nitrate Nitrogen as NO3N	**0.2**	mg/L	0.1	1	10	12/17/23 18:39	EPA 300.0		BEL0662
Hydroxide as CaCO3	ND	mg/L	1.00	1		12/20/23 11:04	SM 2320 B		BEL0726
pH	**7.9**	units	1.0	1		12/20/23 11:04	SM 4500-H+	H	BEL0726
Temperature	**25.0**	units	0.0	1		12/20/23 11:04	SM 4500-H+	H	BEL0726
Sulfate (SO4)	**82.7**	mg/L	0.5	1	250	12/17/23 18:39	EPA 300.0		BEL0662
Total Filterable Solids (TDS)	**530**	mg/L	10.0	1		12/19/23 14:50	SM 2540 C		BEL0670

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Sample Results
(Continued)

Sample: DW - MB
23L1014-02 (Water)

Sampled: 12/13/2023 14:35
Sampled By: M Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Alkalinity as CaCO3	**188**	mg/L	10.0	1		12/20/23 11:09	SM 2320 B		BEL0726
Calcium	**42.8**	mg/L	0.1	1		12/19/23 11:41	EPA 200.7		BEL0695
Chloride	**135**	mg/L	0.2	1	250	12/19/23 02:06	EPA 300.0		BEL0510
Carbonate as CaCO3	ND	mg/L	1	1		12/20/23 11:09	SM 2320 B		BEL0726
Electrical Conductivity	**0.99**	mmhos/cm	0.01	1		12/20/23 11:09	SM 2510 B		BEL0726
Electrical Conductivity umhos	**992**	umhos/cm	10.0	1		12/20/23 11:09	SM 2510 B		BEL0726
Bicarbonate as CaCO3	**188**	mg/L	5.00	1		12/20/23 11:09	SM 2320 B		BEL0726
Potassium	**1.81**	mg/L	0.500	1		12/19/23 11:41	EPA 200.7		BEL0695
Magnesium	**15.8**	mg/L	0.1	1		12/19/23 11:41	EPA 200.7		BEL0695
Sodium	**130**	mg/L	1	1		12/19/23 11:41	EPA 200.7		BEL0695
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 14:35	Field		BEL0654
Nitrate Nitrogen as NO3N	**0.1**	mg/L	0.1	1	10	12/19/23 02:06	EPA 300.0		BEL0510
Hydroxide as CaCO3	ND	mg/L	1.00	1		12/20/23 11:09	SM 2320 B		BEL0726
pH	**7.8**	units	1.0	1		12/20/23 11:09	SM 4500-H+	H	BEL0726
Temperature	**25.0**	units	0.0	1		12/20/23 11:09	SM 4500-H+	H	BEL0726
Sulfate (SO4)	**77.4**	mg/L	0.5	1	250	12/19/23 02:06	EPA 300.0		BEL0510
Total Filterable Solids (TDS)	**600**	mg/L	10.0	1		12/19/23 14:50	SM 2540 C		BEL0670

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0510										
Blank (BEL0510-BLK1)					Prepared & Analyzed: 12/18/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
Blank (BEL0510-BLK2)					Prepared & Analyzed: 12/18/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
Blank (BEL0510-BLK3)					Prepared & Analyzed: 12/19/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
LCS (BEL0510-BS1)					Prepared & Analyzed: 12/18/2023					
Chloride	4.7		0.2	mg/L	5.000		94.4	90-110		
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.4	90-110		
Sulfate (SO4)	4.3		0.5	mg/L	5.000		87.0	90-110		
LCS (BEL0510-BS2)					Prepared & Analyzed: 12/19/2023					
Chloride	4.7		0.2	mg/L	5.000		94.4	90-110		
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.7	90-110		
Sulfate (SO4)	4.4		0.5	mg/L	5.000		87.1	90-110		
Duplicate (BEL0510-DUP1)		**Source: 23L0774-11**			Prepared & Analyzed: 12/18/2023					
Chloride	24.8		0.2	mg/L		24.7			0.469	10
Nitrate Nitrogen as NO3N	0.03		0.1	mg/L		0.03			6.45	10
Sulfate (SO4)	17.5		0.5	mg/L		17.4			0.556	10
Duplicate (BEL0510-DUP2)		**Source: 23L0915-01**			Prepared & Analyzed: 12/19/2023					
Chloride	4.1		0.2	mg/L		4.1			0.610	10
Nitrate Nitrogen as NO3N	7.0		0.1	mg/L		6.9			0.649	10
Sulfate (SO4)	5.8		0.5	mg/L		5.7			0.936	10
Matrix Spike (BEL0510-MS1)		**Source: 23L0774-11**			Prepared & Analyzed: 12/18/2023					
Chloride	29.1		0.2	mg/L	5.000	24.7	89.4	90-110		
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000	0.03	102	90-110		
Sulfate (SO4)	22.4		0.5	mg/L	5.000	17.4	100	90-110		
Matrix Spike (BEL0510-MS2)		**Source: 23L0915-01**			Prepared & Analyzed: 12/19/2023					
Chloride	9.2		0.2	mg/L	5.000	4.1	103	90-110		
Nitrate Nitrogen as NO3N	12.0		0.1	mg/L	5.000	6.9	101	90-110		
Sulfate (SO4)	10.8		0.5	mg/L	5.000	5.7	102	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0510 (Continued)										
Reference (BEL0510-SRM1)					Prepared & Analyzed: 12/18/2023					
Chloride	12.7			mg/L	12.50		101	90-110		
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Sulfate (SO4)	9.7			mg/L	10.00		97.1	90-110		
Reference (BEL0510-SRM2)					Prepared & Analyzed: 12/18/2023					
Chloride	12.7			mg/L	12.50		102	90-110		
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Sulfate (SO4)	9.7			mg/L	10.00		97.2	90-110		
Reference (BEL0510-SRM3)					Prepared & Analyzed: 12/19/2023					
Chloride	12.8			mg/L	12.50		102	90-110		
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Sulfate (SO4)	9.7			mg/L	10.00		97.5	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0662										
Blank (BEL0662-BLK1)					Prepared & Analyzed: 12/16/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
Blank (BEL0662-BLK2)					Prepared & Analyzed: 12/17/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
Blank (BEL0662-BLK3)					Prepared & Analyzed: 12/17/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
Blank (BEL0662-BLK4)					Prepared & Analyzed: 12/17/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
Blank (BEL0662-BLK5)					Prepared & Analyzed: 12/17/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
LCS (BEL0662-BS1)					Prepared & Analyzed: 12/17/2023					
Chloride	4.6		0.2	mg/L	5.000		92.1	90-110		
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000		96.0	90-110		
Sulfate (SO4)	4.4		0.5	mg/L	5.000		87.1	90-110		
LCS (BEL0662-BS2)					Prepared & Analyzed: 12/17/2023					
Chloride	4.7		0.2	mg/L	5.000		93.5	90-110		
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.6	90-110		
Sulfate (SO4)	4.4		0.5	mg/L	5.000		88.7	90-110		
LCS (BEL0662-BS3)					Prepared & Analyzed: 12/17/2023					
Chloride	4.8		0.2	mg/L	5.000		95.2	90-110		
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.0	90-110		
Sulfate (SO4)	4.5		0.5	mg/L	5.000		90.0	90-110		
LCS (BEL0662-BS4)					Prepared & Analyzed: 12/17/2023					
Chloride	4.8		0.2	mg/L	5.000		95.5	90-110		
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.4	90-110		
Sulfate (SO4)	4.5		0.5	mg/L	5.000		90.5	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0662 (Continued)										
Duplicate (BEL0662-DUP1)			**Source: 23L0491-01**		Prepared & Analyzed: 12/17/2023					
Chloride	23.6		0.2	mg/L		23.7			0.587	10
Nitrate Nitrogen as NO3N	0.1		0.1	mg/L		0.1			4.38	10
Sulfate (SO4)	4.1		0.5	mg/L		4.1			1.18	10
Duplicate (BEL0662-DUP2)			**Source: 23L0491-07**		Prepared & Analyzed: 12/17/2023					
Chloride	10.0		0.2	mg/L		9.9			0.331	10
Nitrate Nitrogen as NO3N	1.8		0.1	mg/L		1.8			0.0567	10
Sulfate (SO4)	8.7		0.5	mg/L		8.7			0.334	10
Duplicate (BEL0662-DUP3)			**Source: 23L0991-01**		Prepared & Analyzed: 12/17/2023					
Chloride	47.5		0.2	mg/L		46.7			1.68	10
Nitrate Nitrogen as NO3N	0.3		0.1	mg/L		0.3			0.719	10
Sulfate (SO4)	22.4		0.5	mg/L		22.0			1.84	10
Duplicate (BEL0662-DUP4)			**Source: 23L0991-03**		Prepared & Analyzed: 12/17/2023					
Chloride	48.3		0.2	mg/L		47.9			0.824	10
Nitrate Nitrogen as NO3N	0.3		0.1	mg/L		0.3			0.347	10
Sulfate (SO4)	22.4		0.5	mg/L		22.2			0.730	10
Matrix Spike (BEL0662-MS1)			**Source: 23L0491-01**		Prepared & Analyzed: 12/17/2023					
Chloride	28.0		0.2	mg/L	5.000	23.7	84.6	90-110		
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000	0.1	95.2	90-110		
Sulfate (SO4)	8.8		0.5	mg/L	5.000	4.1	94.4	90-110		
Matrix Spike (BEL0662-MS2)			**Source: 23L0491-07**		Prepared & Analyzed: 12/17/2023					
Chloride	14.7		0.2	mg/L	5.000	9.9	95.7	90-110		
Nitrate Nitrogen as NO3N	6.8		0.1	mg/L	5.000	1.8	100	90-110		
Sulfate (SO4)	13.5		0.5	mg/L	5.000	8.7	96.1	90-110		
Matrix Spike (BEL0662-MS3)			**Source: 23L0991-01**		Prepared & Analyzed: 12/17/2023					
Chloride	51.8		0.2	mg/L	5.000	46.7	101	90-110		
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000	0.3	99.7	90-110		
Sulfate (SO4)	27.1		0.5	mg/L	5.000	22.0	103	90-110		
Matrix Spike (BEL0662-MS4)			**Source: 23L0991-03**		Prepared & Analyzed: 12/17/2023					
Chloride	53.5		0.2	mg/L	5.000	47.9	113	90-110		
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000	0.3	101	90-110		
Sulfate (SO4)	27.7		0.5	mg/L	5.000	22.2	110	90-110		
Reference (BEL0662-SRM1)					Prepared & Analyzed: 12/16/2023					
Chloride	12.3			mg/L	12.50		98.2	90-110		
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		98.9	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0662 (Continued)										
Reference (BEL0662-SRM1)					Prepared & Analyzed: 12/16/2023					
Sulfate (SO4)	9.5			mg/L	10.00		95.2	90-110		
Reference (BEL0662-SRM2)					Prepared & Analyzed: 12/17/2023					
Chloride	12.2			mg/L	12.50		97.8	90-110		
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		98.4	90-110		
Sulfate (SO4)	9.4			mg/L	10.00		94.0	90-110		
Reference (BEL0662-SRM3)					Prepared & Analyzed: 12/17/2023					
Chloride	12.3			mg/L	12.50		98.6	90-110		
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.3	90-110		
Sulfate (SO4)	9.5			mg/L	10.00		95.0	90-110		
Reference (BEL0662-SRM4)					Prepared & Analyzed: 12/17/2023					
Chloride	12.5			mg/L	12.50		99.7	90-110		
Nitrate Nitrogen as NO3N	10.0			mg/L	10.00		100	90-110		
Sulfate (SO4)	9.6			mg/L	10.00		95.8	90-110		
Reference (BEL0662-SRM5)					Prepared & Analyzed: 12/17/2023					
Chloride	12.5			mg/L	12.50		99.9	90-110		
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Sulfate (SO4)	9.6			mg/L	10.00		96.4	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0670										
Blank (BEL0670-BLK1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEL0670-BS1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Total Filterable Solids (TDS)	32.5		10.0	mg/L	2000		1.62	0-200		
Duplicate (BEL0670-DUP1)			**Source: 23L0990-01**		Prepared: 12/18/2023	Analyzed: 12/19/2023				
Total Filterable Solids (TDS)	780		10.0	mg/L		760			2.60	10
Duplicate (BEL0670-DUP2)			**Source: 23L1014-01**		Prepared: 12/18/2023	Analyzed: 12/19/2023				
Total Filterable Solids (TDS)	550		10.0	mg/L		530			3.70	10
Reference (BEL0670-SRM1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Total Filterable Solids (TDS)	397			mg/L	390.0		102	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0695										
Blank (BEL0695-BLK1)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Sodium	ND		1	mg/L						
Calcium	ND		0.1	mg/L						
Potassium	ND		0.500	mg/L						
Magnesium	ND		0.1	mg/L						
Blank (BEL0695-BLK2)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Sodium	ND		1	mg/L						
Potassium	ND		0.500	mg/L						
Calcium	ND		0.1	mg/L						
Magnesium	ND		0.1	mg/L						
LCS (BEL0695-BS1)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Sodium	36		1	mg/L	35.71		101	90-110		
Potassium	35.1		0.500	mg/L	35.71		98.2	90-110		
Calcium	36.0		0.1	mg/L	35.71		101	90-110		
Magnesium	35.4		0.1	mg/L	35.71		99.1	90-110		
LCS (BEL0695-BS2)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Sodium	36		1	mg/L	35.71		102	90-110		
Calcium	36.0		0.1	mg/L	35.71		101	90-110		
Potassium	35.4		0.500	mg/L	35.71		99.1	90-110		
Magnesium	35.6		0.1	mg/L	35.71		99.7	90-110		
Duplicate (BEL0695-DUP1)		**Source: 23L0991-03**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Sodium	76		1	mg/L		75			1.20	15
Calcium	7.2		0.1	mg/L		7.4			1.70	15
Potassium	0.394		0.500	mg/L		0.368			7.01	15
Magnesium	ND		0.1	mg/L		0.1				15
Matrix Spike (BEL0695-MS1)		**Source: 23L0991-03**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Sodium	113		1	mg/L	35.71	75	105	90-110		
Potassium	36.7		0.500	mg/L	35.71	0.368	102	90-110		
Calcium	44.6		0.1	mg/L	35.71	7.4	104	90-110		
Magnesium	36.1		0.1	mg/L	35.71	0.1	101	90-110		
Matrix Spike (BEL0695-MS2)		**Source: 23L1002-06**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Sodium	168		1	mg/L	35.71	138	83.2	90-110		
Potassium	37.1		0.500	mg/L	35.71	0.926	101	90-110		
Calcium	87.9		0.1	mg/L	35.71	53.5	96.4	90-110		
Magnesium	72.8		0.1	mg/L	35.71	39.3	93.9	90-110		
Reference (BEL0695-SRM2)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Potassium	20.5			mg/L	21.90		93.7	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0695 (Continued)										
Reference (BEL0695-SRM2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Sodium	92			mg/L	91.50		101	90-110		
Reference (BEL0695-SRM3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Calcium	46.3			mg/L	45.90		101	90-110		
Magnesium	35.0			mg/L	35.60		98.4	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0726										
Blank (BEL0726-BLK1)					Prepared: 12/19/2023 Analyzed: 12/20/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Hydroxide as CaCO3	ND		1.00	mg/L						
pH	5.1		1.0	units						
Alkalinity as CaCO3	ND		10.0	mg/L						
Carbonate as CaCO3	ND		1	mg/L						
Temperature	25.0		0.0	units						
Bicarbonate as CaCO3	ND		5.00	mg/L						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0726-BLK2)					Prepared: 12/19/2023 Analyzed: 12/20/2023					
Alkalinity as CaCO3	ND		10.0	mg/L						
Carbonate as CaCO3	ND		1	mg/L						
Electrical Conductivity	ND		0.01	mmhos/cm						
Hydroxide as CaCO3	ND		1.00	mg/L						
pH	5.4		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	units						
Bicarbonate as CaCO3	ND		5.00	mg/L						
Duplicate (BEL0726-DUP1)		**Source: 23L1085-04**			Prepared: 12/19/2023 Analyzed: 12/20/2023					
Alkalinity as CaCO3	236		10.0	mg/L		240			1.74	10
Carbonate as CaCO3	ND		1	mg/L		ND				10
pH	7.8		1.0	units		7.7			0.388	10
Electrical Conductivity	0.67		0.01	mmhos/cm		0.67			0.237	10
Hydroxide as CaCO3	ND		1.00	mg/L		ND				10
Electrical Conductivity umhos	675		10.0	umhos/cm		673			0.237	10
Duplicate (BEL0726-DUP2)		**Source: 23L1094-01**			Prepared: 12/19/2023 Analyzed: 12/20/2023					
Hydroxide as CaCO3	ND		1.00	mg/L		ND				10
Carbonate as CaCO3	ND		1	mg/L		ND				10
pH	7.1		1.0	units		7.1			0.281	10
Electrical Conductivity	0.44		0.01	mmhos/cm		0.44			0.797	10
Alkalinity as CaCO3	117		10.0	mg/L		118			0.742	10
Electrical Conductivity umhos	441		10.0	umhos/cm		438			0.797	10
Reference (BEL0726-SRM1)					Prepared: 12/19/2023 Analyzed: 12/20/2023					
Alkalinity as CaCO3	119			mg/L	128.0		92.8	90-110		
Electrical Conductivity	440			umhos/cm	426.0		103	90-110		
Reference (BEL0726-SRM2)					Prepared: 12/19/2023 Analyzed: 12/20/2023					
Alkalinity as CaCO3	126			mg/L	128.0		98.5	90-110		
Electrical Conductivity	436			umhos/cm	426.0		102	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

M & A Dairy #1
14720 Carlucci Rd
Dos Palos, CA 93620

Account# 00-0021357
Account Manager: Ben Nydam
Submitted By: Aurelia Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:32

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0726 (Continued)										
Reference (BEL0726-SRM3)					Prepared: 12/19/2023	Analyzed: 12/20/2023				
Electrical Conductivity	449			umhos/cm	426.0		105	90-110		
Alkalinity as CaCO3	130			mg/L	128.0		101	90-110		
Reference (BEL0726-SRM4)					Prepared: 12/19/2023	Analyzed: 12/20/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0726-SRM5)					Prepared: 12/19/2023	Analyzed: 12/20/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0726-SRM6)					Prepared: 12/19/2023	Analyzed: 12/20/2023				
pH	4.0			units	4.000		99.8	97.5-102.5		
Reference (BEL0726-SRM7)					Prepared: 12/19/2023	Analyzed: 12/20/2023				
pH	7.5			units	7.520		99.9	67021-101.32		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

12/14/23 14:20 23L1014

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Acct No. / Cons.
Bill To: 21357 / 8

Purchase Order No. _____ Results Needed By _____

Client: **M & A Dairy #1**
Address: 14651 S Carlucci Road
City, State, Zip: Dos Palos, CA 93620
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Aurelia Goncalves - (209) 387-4313

Ranch _____

Date sampled: 12/13/23

Sampled by: MPed

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: 2 No. Bottles: 2

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

( ) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

(X) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-Shop	Sampled From:	12/13/23	230P	0	4.7
2. DW-MB	Sampled From:	12/13/23	235P	0	~~5.0~~ 4.3 JG
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MPed	CAS Inc	12/13/23 235P	12/14/23 12p
Second	CSM	DLI	12/14/23 12pm	12/14/23 2:15pm
Third				
Fourth	[illegible]	DLI	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Shipping
$ _____ In
$ _____ Out

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid ___ Rec By ___ Check No. ___ Date ___

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/14/23 14:20 23L1014

SEND OUT